

FINITE-USE DISCRIMINATION GEOMETRY OF ORDERED QUANTUM MEASUREMENTS

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ABSTRACT. We study finite decision experiments on ordered quantum measurements when each fresh acquisition is conditionally independent of the retained receiver. For arbitrary finite two-call experiments we give an attained common-barycenter variational principle and a Hermitian-majorant dual, including specified retained-register dimensions. A Heisenberg-picture postponement lemma identifies the value without a receiver message. With the complete old Schmidt support retained, messages are useless at every prescribed pure first acquisition exactly when each labelwise continuation value is concave. If this condition fails, a two-outcome receiver instrument after one device label witnesses an advantage at some first acquisition. For a weighted shared-basis experiment with a specified biased prior, we evaluate every prescribed-initial-spectrum value through the least rank-capped majorant and a secular equation. An instrument with at most d outcomes attains it without changing the initial acquisition. We obtain its exact message-gain criterion, saturation and rigidity. For equal priors and a two-dimensional old receiver we also give the full initial-spectrum formula in every input dimension, including the complete qubit case. The arbitrary-experiment characterization and the shared-basis closed formulas have different quantifiers. The previously established local tangent-tube, profile and hard-budget geometry is retained in the linked supplement.

1. INTRODUCTION

A quantum measurement can produce only a classical label while leaving information in a reference prepared by the experimenter. Across two uses, retaining this receiver, sending its classical measurement result to a fresh source, and coherently feeding it into a later acquisition are different operations. This article studies their exact decision values in a fixed operational interface.

An ordered measurement is a tuple $E = (E_y)_y$ of positive matrices with $\sum_y E_y = I$. Its unknown interface is $\mathcal{M}_E(X) = \sum_y \text{tr}(E_y X) |y\rangle\langle y|$; no residual device output is accessible. A reset cut requires the *entire* new input–reference system to be conditionally independent of the old receiver. The old and fresh references may be measured jointly only after the second call. All controls are common to the hypotheses, and all formal histories carry a defined rule. We use unhalved trace norms; a decision-probability perturbation carries the corresponding factor $1/2$ for a trace-zero difference.

1.1. General experiments and the value of a message. For arbitrary finite priors and rewards, the branch normal form is $\sum_h A_{yh} = \rho$, with a fresh density b_{yh} on each branch. The trace of A_{yh} is a Gram-decomposition weight, not a physical history probability. Optimizing the fresh state and the final POVM gives a continuation function g_y ; receiver messages replace it by its upper concave envelope c_y . Theorems 2.4 and 2.5 give attained primal and dual values, with finite instruments and universal Hermitian majorants. Their rank-resolved forms specify which complete quantum registers are counted at each cut.

Lemma 3.1 proves the no-message reduction directly in the Heisenberg picture. Theorem 3.2 makes its uniform consequence exact: with the full old Schmidt support retained, receiver messages have no value at every prescribed pure first acquisition if and only if every fresh-rank-constrained

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continuation is concave. Failure is witnessed by two outcomes after one first device label at some initial acquisition. This assertion is not a two-message bound at every preassigned initial state. Nor may postponement bypass an old-dimension bound smaller than the system it retains.

1.2. A solved finite family. We next use a weighted exact second-moment family of ordered bases. An alternative basis is selected once and reused at both calls; independent redraws give a different experiment with no second-moment signal. Theorem 4.2 evaluates the three unrestricted endpoint classes, and Theorem 6.5 resolves both retained-register dimensions.

For the specified biased prior and a prescribed pure initial Gram ρ , Theorem 8.2 gives

$$P_{k,\ell}^b(\rho) = \frac{d+1}{2(d+1)-t^2} + \frac{t^2}{2[2(d+1)-t^2]} \chi(q^{(\min(k,\ell))}(\lambda(\rho))).$$

The boundary-safe definitions of χ and the unique least rank-capped majorant $q^{(r)}$, including ties, are given before that theorem in Section 8. A finite instrument on the fixed initial receiver attains the formula. No-message preparation instead uses the leading truncated spectrum. Thus at $t > 0$ the message has strict value exactly for $2 \leq r < \text{rank } \rho$. Theorem 7.1 and the subsequent rigidity estimates describe equality and quantitative loss. Proposition 9.2 works out a complete three-dimensional example, with exact algebraic values and rational secular intervals.

The prior matters. For equal priors and old receiver dimension two, Theorem 10.1 gives a different initial-spectrum formula and different optimal fresh Schmidt weights. Its qubit specialization covers every pure first acquisition without a receiver message. The full proof includes the optimizing fresh weights, the singular boundary and finite attainment on the prescribed receiver.

1.3. Relation to local geometry. The same ordered-measurement interface also supports local fixed-tube results in the linked supplement. Those results classify support and covariance directions, preserve the finite tube remainders, and give allocation orders involving $N = \sum_j n_j$, $V = \sum_j n_j^2$, and hard pathwise Q . They are not used to prove the exact finite-game theorems. Conversely the spectral formulas do not make their fixed-object constants uniform over changing support strata. The local theory is retained as mathematics, not discarded; it is separated from the finite-experiment proof route. The independent structural article and the wider analytic Theta programme supply no hidden hypotheses here.

2. A VARIATIONAL PRINCIPLE FOR TWO-CALL RESET EXPERIMENTS

The fresh-preparation cut admits an exact finite-dimensional description of optimal decision values. No symmetry, second-moment identity, or special choice of prior is needed in this section. The description is by a concave envelope over density matrices, not over pure states alone. Its finite realization bounds the receiver needed to attain a value; it does not identify reset width with a prescribed memory dimension.

2.1. Experiments and strategy classes. Let $I_\ell = \mathbb{C}^{d_\ell}$ and let Y_ℓ be a finite classical alphabet, $\ell = 1, 2$. A finite experiment consists of a probability law (p_θ) , two ordered measurements $E^{\theta,\ell}$ on I_ℓ , and rewards $r_{j\theta} \in [0, 1]$ for decisions $1 \leq j \leq m$. The index θ is sampled once. Conditional on it the two slots are the memoryless measurement channels $\mathcal{M}_{E^{\theta,\ell}}$. Repeating one unknown measurement is the case $E^{\theta,1} = E^{\theta,2}$. Identification and binary decisions on subfamilies are obtained by taking indicator rewards.

Definition 2.1 (Two-call classes). An unrestricted strategy prepares a state on $I_1 \otimes R$, observes the first label y , applies a common trace-preserving map from R to $I_2 \otimes R'$, and, after observing z , measures R' to decide j . A unit-reset strategy instead applies a normal instrument $(\mathcal{K}_{yh})_h$ to R , retains its old quantum output O_{yh} , and prepares a state σ_{yh} on the complete fresh system $I_2 \otimes R_{2,yh}$. Conditional on (y, h) the preparation is exactly the tensor product of the old output

with σ_{yh} . No old quantum register enters I_2 or its fresh reference. After the second label, the two receivers may be measured jointly. A classically adaptive strategy is a unit-reset strategy that measures the entire old receiver and retains only a classical record before fresh preparation.

All rules are independent of θ and defined on every formal history, including histories null under a hypothesis. Receiver spaces have no dimension bound. Instruments may have countably many outcomes; all classical records are retained. Finite or countable public randomization and the finite-dimensional trace-norm closure of the resulting output-dephased Choi tester tuples are included in each class. General public seed laws are included through this closure: their finitely many tester coordinates are limits of finite convex combinations. Early stopping is implemented by a fixed, discarded fresh call and a decision independent of its label; it is not a zero-length component of a prescribed profile. Write $P_{\text{all}}, P_{\text{reset}}, P_{\text{ca}}$ for the suprema of expected reward in these classes.

This reset cut is the two-call instance of Definition S12.1. The first class in the ordering $P_{\text{ca}} \leq P_{\text{reset}} \leq P_{\text{all}}$ agrees with complete measure-and-reprepare adaptation, not with a constraint of separability on all final reset effects. In the order I_1, Y_1, I_2, Y_2 , a classically adaptive effect is a sum of positive products across $I_1 Y_1 \mid I_2 Y_2$ [2, Definition 23]. The positive separable cone is closed in these finite dimensions: its trace-one slice is the compact convex hull of product states, and trace controls passage to the cone. Only this necessary condition, not its sufficiency for classical adaptive normalization, is used below.

Lemma 2.2 (Countable records). *Let (X_h) be positive trace-class operators with $\sum_h \text{tr } X_h < \infty$. Their partial sums converge in trace norm, and their recorded direct sum lies in the trace-class ℓ^1 sum. For effects $0 \preceq F_h \preceq I$, the corresponding probability series converges absolutely. Replacing all but finitely many branches of a normal instrument by one remainder branch preserves normalization; changing the continuation on that branch changes any reward in $[0, 1]$ by at most its probability.*

Proof. The norm of a positive tail equals its trace. This proves the first assertions and bounds the probability tail by $\sum_{h>n} \text{tr } X_h$. For an instrument, the sum of the omitted completely positive maps is the remainder map; its trace is precisely the omitted outcome probability. Apply the bound to each member of the finite experiment and then average. The maximum of its finitely many tail probabilities tends to zero. No conditional state is divided by a history probability. $\square$

2.2. A complete branch normal form. Fix bases on I_1, I_2 . A coefficient map $C : I_1 \rightarrow R$ represents $|C\rangle = \sum_i |i\rangle \otimes C|i\rangle$; its squared norm is $\text{tr } C^* C$. Measuring its input gives the unnormalized receiver state $C E_y^\top C^*$.

Lemma 2.3 (Unit-reset normal form and converse). *Every implemented unit-reset strategy has a purification, still respecting the reset cut, from which its original decision statistics are recoverable by common receiver channels. For this purification there are rectangular maps*

$$C_0 : I_1 \longrightarrow R_0, \quad C_{yh} : I_1 \longrightarrow O_{yh}, \quad D_{yh} : I_2 \longrightarrow R_{2,yh}$$

with, on the input copies,

$$(2.1) \quad \rho = C_0^* C_0, \quad \text{tr } \rho = 1, \quad A_{yh} = C_{yh}^* C_{yh}, \quad \sum_h A_{yh} = \rho, \quad b_{yh} = D_{yh}^* D_{yh}, \quad \text{tr } b_{yh} = 1.$$

The receiver block on the formal record (y, h, z) is

$$(2.2) \quad (C_{yh} E_y^{\theta,1 \top} C_{yh}^*) \otimes (D_{yh} E_z^{\theta,2 \top} D_{yh}^*).$$

The sum in (2.1) converges in trace norm. Conversely, every finite collection satisfying (2.1) is realizable, up to common receiver isometries, by a unit-reset strategy with old receiver dimension at most d_1 and fresh reference dimension at most d_2 . Zero-probability histories require no extra

assumption. The tester closure in Definition 2.1 is used by continuity of decision values; no physical dilation is assumed for an abstract limit before its value has been realized by the finite optimization.

Proof. Purify the first preparation, retaining the purifying system in R_0 . The Schmidt support has dimension at most d_1 . For each y , dilate each branch map of the intervening instrument to a single map $V_{yh} : R_0 \rightarrow O_{yh}$, retaining its environment as old receiver memory. On the relevant support, $\sum_h V_{yh}^* V_{yh} = I$ in the strong sense. Thus $C_{yh} = V_{yh} C_0$ has $\sum_h C_{yh}^* C_{yh} = C_0^* C_0$. The positive tails converge in trace norm by Lemma 2.2. This argument uses only the finite Schmidt support even when the original receiver or a dilation environment is infinite-dimensional. Tracing those environments recovers the original instrument.

Purify the entire fresh preparation separately. Its coefficient map is D_{yh} with squared Hilbert–Schmidt norm one. The tensor-product preparation rule, not a statement about a probe marginal, gives (2.2). Common pre-query controls are absorbed into these preparations and common post-query controls into the final receiver measurement. Fresh mixing may be purified inside the fresh reference. A retained public seed may be put in the classical record; randomization before the first call is a mixture of such strategies and cannot increase the supremum of a linear reward. More explicitly, for seed law q_s and first coefficient maps $C_{0,s}$, take $C_0 = \bigoplus_s \sqrt{q_s} C_{0,s}$ and measure the seed register as part of the first receiver instrument. The branch maps are $C_{y,(s,h)} = \sqrt{q_s} V_{ysh} C_{0,s}$, and their Gram sum is $\sum_s q_s C_{0,s}^* C_{0,s} = C_0^* C_0$ for every y . The fresh rule retains its dependence on (s, y, h) , entirely on the fresh side of the cut. Thus the original seeded decision statistics are recovered. Countable records and tester limits follow from the preceding lemma and continuity.

For the converse let $r = \text{rank } \rho$ and choose $C_0 : I_1 \rightarrow \mathbb{C}^r$ with $C_0^* C_0 = \rho$. Its restriction to $\text{supp } \rho$ is invertible onto $\mathbb{C}^r$. Let C_0^+ be the inverse on this support, and set

$$C_{yh} = \sqrt{A_{yh}}, \quad V_{yh} = \sqrt{A_{yh}} C_0^+, \quad D_{yh} = \sqrt{b_{yh}}.$$

Positivity and $\sum_h A_{yh} = \rho$ imply $\text{supp } A_{yh} \subseteq \text{supp } \rho$. Hence $V_{yh} C_0 = \sqrt{A_{yh}}$ and $\sum_h V_{yh}^* V_{yh} = I_r$. These are Kraus operators of a trace-preserving recorded instrument with outputs in $\mathbb{C}^{d_1}$. The fresh vector $|\sqrt{b_{yh}}\rangle$ is normalized and independent of that output. Polar decomposition identifies any other rectangular maps of the same Gram matrices with these maps on their supports; receiver isometries may be absorbed into the final effects. Zero Gram matrices give zero branch maps. All other null histories are handled by the same unnormalized formulas, including projective devices. $\square$

2.3. The compact primal problem. Let $\mathcal{D}_d$ denote the density matrices on $\mathbb{C}^d$, and put

$$(2.3) \quad G_{j,yz} = \sum_{\theta} p_{\theta} r_{j\theta} (E_y^{\theta,1})^{\top} \otimes (E_z^{\theta,2})^{\top}.$$

These are fixed positive matrices on the two input copies. For $A \succeq 0$ and $b \in \mathcal{D}_{d_2}$ define

$$(2.4) \quad \Phi_y(A, b) = \sum_z \max_{\substack{F_j \succeq 0 \\ \sum_j F_j = I}} \sum_j \text{tr}[F_j(\sqrt{A} \otimes \sqrt{b}) G_{j,yz}(\sqrt{A} \otimes \sqrt{b})].$$

The POVM is chosen separately at every recorded z . It acts on $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$. For $a \in \mathcal{D}_{d_1}$ let $g_y(a) = \max_{b \in \mathcal{D}_{d_2}} \Phi_y(a, b)$ and let

$$(2.5) \quad c_y(\rho) = \sup \left\{ \sum_h \lambda_h g_y(a_h) : \lambda_h \geq 0, \sum_h \lambda_h = 1, \sum_h \lambda_h a_h = \rho \right\}.$$

The envelope is taken over *all* density matrices a_h . Replacing them by rank-one states would change the resource problem. The construction of concave envelopes is standard [7]; the operational content here is the reset realization and the common barycenter constraint across all first labels.

Theorem 2.4 (Exact reset variational principle). *For every finite experiment of Definition 2.1,*

$$(2.6) \quad P_{\text{reset}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y(\rho) = \max_{\substack{A_{yh} \geq 0, \sum_h A_{yh} = \rho \\ \rho \in \mathcal{D}_{d_1}, b_{yh} \in \mathcal{D}_{d_2}}} \sum_{y,h} \Phi_y(A_{yh}, b_{yh}).$$

All maxima are attained. For a fixed ρ of rank r , at most r^2 nonzero instrument outcomes are required after each first label. An optimal strategy exists with old receiver dimension at most d_1 , fresh reference dimension at most d_2 , and a final quantum receiver of dimension at most $d_1 d_2$, separately from the finite classical record. The dimension bounds concern the retained registers at the recording cuts, not temporary workspaces for implementing an instrument. These assertions include arbitrary priors and arbitrary fixed pairs as particular decision problems; they assert an exact optimum, not a strict advantage for every such problem.

Proof. For fixed y , Φ_y is continuous, and is positively homogeneous in A . Continuity follows from continuity of matrix square roots and maximization of a continuous function over a compact POVM set. Maximizing also over compact $\mathcal{D}_{d_2}$ makes g_y continuous. The graph $\{(a, g_y(a)) : a \in \mathcal{D}_{d_1}\}$ is compact in a real space of affine dimension at most d_1^2 . Its convex hull is compact. The largest height above ρ is therefore attained and is exactly $c_y(\rho)$, also allowing countable decompositions. It is an upper semicontinuous concave function of ρ . Consequently $\sum_y c_y$ attains a maximum.

Carathéodory first supplies at most $r^2 + 1$ atoms at a maximizing height over a rank- r matrix: each atom is supported on $\text{supp } \rho$, since positive matrices cannot cancel on its orthogonal complement. If more than r^2 weights are nonzero, the Hermitian atoms are linearly dependent. Choose nonzero real v_h with $\sum_h v_h a_h = 0$; tracing gives $\sum_h v_h = 0$. Small perturbations $\lambda_h \pm \epsilon v_h$ preserve the barycenter and positivity. Optimality forces $\sum_h v_h g_y(a_h) = 0$, since otherwise one sign raises the height. Move in this direction until a weight vanishes. The value is unchanged. Iteration gives at most r^2 atoms. For each atom the maximum over b and the leaf POVMs is attained.

By Lemma 2.3, the value of any purified competitor is bounded by the right side of (2.6), with $A_{yh} = \lambda_{yh} a_{yh}$. Countable records follow by the positive tail bound; public mixtures and norm limits preserve the upper. Conversely, the lemma realizes every finite maximizing collection, and its optimal leaf POVMs realize every maximum in (2.4). This proves equality and the dimension bounds. The final quantum dimension excludes the separately recorded labels. $\square$

For a binary equal-prior fixed pair, its optimal unhalved final trace separation is $4P_{\text{reset}} - 2$. Thus the theorem is also a general variational formula for that operational distance. The primal has finitely many bounded matrix variables, but is not asserted to be a convex semidefinite program or an efficient synthesis algorithm.

2.4. Dual certificates and the role of receiver feedback.

Theorem 2.5 (Duality and contact conditions). *For the same experiment,*

$$(2.7) \quad P_{\text{reset}} = \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{D}_{d_1}),$$

where H_y are Hermitian $d_1 \times d_1$ matrices. The minimum is attained. It is equivalent to minimizing λ subject to the same majorizations and $\sum_y H_y = \lambda I$. For any optimal primal and dual, every atom of positive weight is a contact point, $\text{tr}(H_y a_{yh}) = g_y(a_{yh})$, and ρ is supported on the maximal eigenspace of $\sum_y H_y$. Conversely these conditions, with maximizing fresh states and leaf POVMs, certify optimality.

Proof. Every majorization of g_y majorizes its concave envelope. Therefore $\sum_y c_y(\rho) \leq \text{tr}(\rho \sum_y H_y) \leq \lambda_{\max}(\sum_y H_y)$, which proves weak duality. For completeness, the strong-duality step

is finite-dimensional. Extend c_y homogeneously to positive matrices by $\hat{c}_y(A) = \text{tr } A c_y(A/\text{tr } A)$ and $\hat{c}_y(0) = 0$. Its hypograph

$$\mathcal{K}_y = \{(A, u) : A \succeq 0, u \leq \hat{c}_y(A)\}$$

is a closed convex cone. Closedness at $A = 0$ follows from boundedness of g_y ; elsewhere it follows from the compact-graph construction. The conic problem maximizing $\sum_y u_y$, with $(A_y, u_y) \in \mathcal{K}_y$, $A_y = \rho$ and $\text{tr } \rho = 1$, is precisely the primal problem. The point $A_y = \rho = I/d_1$, $u_y = -1$ is strictly interior to all hypographs, since rewards are nonnegative and A_y is positive definite. Its value is finite, in fact at most one by the operational realization. The finite-dimensional separating-hyperplane form of conic Slater duality thus gives an attained dual with no gap [8, Sections 5.2.3 and 5.9.1].

Its multipliers can be read directly: the supremum of $u - \text{tr}(H_y A)$ on $\mathcal{K}_y$ is zero exactly when $\text{tr}(H_y a) \geq g_y(a)$ for every density matrix. Taking ρ as a free Hermitian variable under its trace constraint imposes $\sum_y H_y = \lambda I$. This gives the stated dual. Conversely, for majorants with sum S , add the positive matrix $\lambda_{\max}(S)I - S$ to one H_y ; all majorizations remain valid and the sum becomes scalar. The two forms have the same minimum. Finally, the weak-duality difference is a sum of nonnegative atom majorization defects and a nonnegative spectral defect. It vanishes exactly under the stated contact and support conditions. $\square$

The majorizations in (2.7) are universal inequalities on density matrices. A finite sampling of them is not a dual certificate.

Corollary 2.6 (Quantitative contact certificates). *Let (H_y) be any feasible all-density majorants in Theorem 2.5, put $S_H = \sum_y H_y$ and $U = \lambda_{\max}(S_H)$, and consider a finite or countable reset normal form with achieved reward v . For every nonzero branch write $w_{yh} = \text{tr } A_{yh}$, $a_{yh} = A_{yh}/w_{yh}$, and let v_{yh} be its contribution to the reward, summed over the second labels and decisions. Then*

$$\begin{aligned} (2.8) \quad U - v &= \text{tr}[\rho(UI - S_H)] \\ &+ \sum_{y,h} w_{yh} [\text{tr}(H_y a_{yh}) - g_y(a_{yh})] \\ &+ \sum_{y,h} [w_{yh} g_y(a_{yh}) - v_{yh}]. \end{aligned}$$

Every term is nonnegative. Consequently, if $U - v \leq \varepsilon$, the total Gram weight of branches with majorization defect at least $u > 0$ is at most ε/u . If the maximal eigenspace projection P of S_H has a spectral gap $\gamma > 0$ to its complement, then $\text{tr}[(I - P)\rho] \leq \varepsilon/\gamma$.

Proof. The common-barycenter constraints give $\sum_{y,h} w_{yh} \text{tr}(H_y a_{yh}) = \text{tr}(\rho S_H)$. Add and subtract this quantity and $\sum_{y,h} w_{yh} g_y(a_{yh})$. The leaf optimization bounds v_{yh} by $w_{yh} g_y(a_{yh})$. Feasible majorization and $S_H \preceq UI$ prove nonnegativity. All series converge absolutely: the first-label alphabet is finite, $\sum_h w_{yh} = 1$ for every y , and the functions and matrices are bounded. Restricting a nonnegative sum proves the first estimate; $UI - S_H \succeq \gamma(I - P)$ proves the second. $\square$

The weights in this certificate are input Gram weights, not the hypothesis-dependent probabilities of recorded histories. With an attained optimal dual, $U - v$ is precisely the loss in optimal reward. The formula requires a genuine universal majorant, not sampled tests of its inequality.

Corollary 2.7 (Label feedback versus receiver feedback). *If no receiver-instrument outcome is used to choose the second preparation, but the first device label may be used, the optimum is*

$$P_{\text{label}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y g_y(\rho).$$

Allowing receiver feedback replaces g_y by c_y in this formula. In particular, concavity of every g_y is sufficient for $P_{\text{label}} = P_{\text{reset}}$. More generally, a label-only strategy with a common density ρ is optimal for the full reset class if it attains the contact and spectral conditions of Theorem 2.5.

Proof. With one preparation choice after each y , the normal form has $A_y = \rho$; a trace-preserving receiver operation cannot improve a subsequent unrestricted joint measurement and can be postponed to it. Conversely these choices are admissible. The assertion follows from the primal formula and the definition of concave envelope. $\square$

2.5. A geometric criterion for retained receiver advantage. Let $\mathcal{P}_{d_1}$ be the rank-one density matrices. In parallel with (2.5), define

$$(2.9) \quad c_y^{\text{cl}}(\rho) = \sup \left\{ \sum_h \lambda_h g_y(a_h) : a_h \in \mathcal{P}_{d_1}, \quad \lambda_h \geq 0, \quad \sum_h \lambda_h a_h = \rho \right\}.$$

The weights sum to one by taking traces. Every density matrix admits such a decomposition, but the value of this pure-atom envelope need not equal the all-density envelope c_y .

Theorem 2.8 (Classical and quantum receiver envelopes). *For every finite experiment of Definition 2.1,*

$$(2.10) \quad P_{\text{ca}} = \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y^{\text{cl}}(\rho) = \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{P}_{d_1}).$$

Both optima are attained. At most $(\text{rank } \rho)^2$ nonzero first-receiver outcomes per first device label suffice at an optimal ρ . The passage from complete classical readout to retained quantum receiver memory replaces the pure-atom envelopes c_y^{cl} by the all-density envelopes c_y . In particular, the invariant

$$(2.11) \quad \Gamma = \max_{\rho} \sum_y c_y(\rho) - \max_{\rho} \sum_y c_y^{\text{cl}}(\rho)$$

is exactly the optimal retained-receiver advantage in that experiment.

Proof. After complete first readout, a classical branch has a positive input Gram effect A and retains no quantum system depending on θ . Its probability under a first effect is $\text{tr}(AE_y^{\text{T}})$. Refine a spectral decomposition $A = \sum_i \lambda_i a_i$ into the classical record and keep the former second preparation and decision on each refined branch. This does not change the score and gives a rank-one decomposition with the same barycenter. Optimizing the second action separately after the refined record can only increase it.

Conversely, for a rank-one atom $a = |u\rangle\langle u|$, the old receiver in the canonical normal form has

$$\sqrt{a} E_y^{\text{T}} \sqrt{a} = \langle u, E_y^{\text{T}} u \rangle a.$$

Its normalized state is independent of the unknown device. A joint POVM with this known factor is equivalent to a POVM on the fresh reference alone, by taking its expectation against a . Thus $g_y(a)$ is attainable with only a classical first record. The converse part of Lemma 2.3, followed by discarding this known old state, realizes every finite pure-atom decomposition. These two observations give the first equality in (2.10).

The graph of g_y on the compact pure-state set is compact. The compact-convex-hull and rank-face atom-elimination arguments of Theorem 2.4 therefore apply verbatim, giving attainment and the stated branch count. Its homogeneous hypograph is again a closed convex cone, with a strictly feasible point at a positive definite common barycenter and negative heights. The proof of Theorem 2.5 applies with majorization required only on pure atoms and gives the attained dual. Majorization on pure atoms is equivalent to majorization of c_y^{cl} on every density matrix, not to

majorization of g_y on every density matrix. The last assertion now follows from the two exact primal formulas. $\square$

Corollary 2.9 (Complete separation certificates). *There is a strict retained-receiver advantage if and only if there exist a finite feasible reset collection from (2.6), with achieved reward v , and Hermitian matrices H_y such that*

$$\mathrm{tr}(H_y a) \geq g_y(a) \quad (a \in \mathcal{P}_{d_1}), \quad v > \lambda_{\max} \left(\sum_y H_y \right).$$

There is no advantage if and only if an attaining classical strategy has contact and spectral equality with an all-density dual majorant of Theorem 2.5.

Proof. The displayed inequalities compare a physically attained reset lower with a valid classical upper. If the gap is positive, take an optimal finite reset collection and an optimal classical dual, whose existence was proved above. This proves both directions of the first assertion. If the gap is zero, an optimal classical strategy also attains the reset value; complementarity with any attained reset dual gives the second condition. Conversely, that condition certifies a common upper and lower for the two classes, so their values agree. $\square$

These are certificates with finitely many matrix variables and universal majorization requirements, not finite-sampling tests. The criterion applies to any fixed-pair problem as well as to ensemble decisions, but does not require every experiment to have positive Γ . For example, the equal-prior qubit calculation below gives pure-atom majorants $H_0 = H_1 = (7/24)I$, whereas a reset strategy attains $5/8$. The gap is $1/24$. Its universal pure-atom majorization follows from the classical upper in Theorem 5.2, not from the finite arithmetic regression.

2.6. The unrestricted causal normalization.

Proposition 2.10 (Physical two-call tester blocks). *In the unnormalized convention $J^\sharp(E) = \sum_y E_y \otimes |y\rangle\langle y|$, every unrestricted strategy has positive blocks $T_{j,yz}$ and positive matrices Ξ_y with*

$$(2.12) \quad \sum_j T_{j,yz} = \Xi_y, \quad \mathrm{tr}_{I_2} \Xi_y = \rho_1, \quad \mathrm{tr} \rho_1 = 1.$$

Its probability of decision j on the output block (y, z) is $\mathrm{tr}[T_{j,yz}(E_y^{\theta,1} \otimes E_z^{\theta,2})]$. In particular $0 \preceq T_{j,yz} \preceq \Xi_y$ and $\mathrm{tr} \Xi_y = 1$.

Proof. Purify the first preparation to $|\psi\rangle \in I_1 \otimes R$. For its common intervening channel $\mathcal{A}_y : R \rightarrow I_2 \otimes R'$ put

$$X_y = (\mathrm{Id} \otimes \mathcal{A}_y)(|\psi\rangle\langle\psi|), \quad \Xi_y = \mathrm{tr}_{R'} X_y, \quad \rho_1 = \mathrm{tr}_R |\psi\rangle\langle\psi|.$$

For the final POVM $(F_{j,yz})_j$ on R' define

$$T_{j,yz} = \mathrm{tr}_{R'} [(I \otimes \sqrt{F_{j,yz}}) X_y (I \otimes \sqrt{F_{j,yz}})].$$

Positivity is immediate. Cyclicity in the traced factor and $\sum_j F_{j,yz} = I$ give the sum; trace preservation of $\mathcal{A}_y$ gives the marginal. Expanding the Born rule gives the contraction. Here ρ_1 is the physical input marginal, whereas the coefficient Gram matrix ρ in (2.1) is its transpose. These two conventions must not be interchanged. Every complementary effect is positive by the same construction. No factor $d_1 d_2$ occurs, because $J^\sharp$ uses the unnormalized maximally entangled vector. Countable records and limits follow by continuity. $\square$

3. RECEIVER MESSAGES AND CONCAVIFICATION

The distinction between a device label and a receiver message is operational, not merely terminological. A device label may select a fresh source even when an intermediate receiver result may not. We first give the exact postponement statement, then characterize when this additional communication has value in an arbitrary finite experiment. The old receiver is large enough to retain the complete initial Schmidt support throughout this section. Only the fresh reference is dimension constrained.

3.1. Postponement without a receiver message. Let R be the initial receiver, I_2 the second input, and F_y the fresh reference after the first device label y . The prescribed fresh state σ_y on $I_2 \otimes F_y$ is independent of a receiver outcome h . A normal instrument $\mathcal{K}_{h|y} : \mathcal{T}(R) \rightarrow \mathcal{T}(O_{yh})$ acts on the old receiver. Write $M_{j|yhz}$ for the final decision POVM on $O_{yh} \otimes F_y$. The record (y, h, z) is classical; retaining h for the final readout is permitted even though it is not sent to the source.

Lemma 3.1 (No-message postponement). *For every such instrument and readout, the operators*

$$(3.1) \quad G_{j|yz} = \sum_h (\mathcal{K}_{h|y}^* \otimes \text{Id}_{F_y})(M_{j|yhz})$$

form a POVM on $R \otimes F_y$. Replacing the instrument and its record by this final POVM preserves every decision probability, under each hypothesis separately, for every input state and every second device. The statement includes countable normal instruments and their finite-dimensional tester closure. It preserves any fresh-reference dimension bound. It preserves an old-receiver bound k whenever $\dim R \leq k$, or after compression to a common input support of dimension at most k .

Proof. Every summand in (3.1) is positive. Since the sum of the instrument maps is trace preserving,

$$\sum_j G_{j|yz} = \sum_h \mathcal{K}_{h|y}^*(I_{O_{yh}}) \otimes I_{F_y} = I_R \otimes I_{F_y}.$$

Let $X_{\theta,y}$ be an arbitrary unnormalized first receiver block and let $Z_{\theta,y,z}$ be the fresh reference block obtained by applying the second device to σ_y . Conditional independence at the preparation cut gives

$$\begin{aligned} \sum_h \text{tr}[M_{j|yhz}(\mathcal{K}_{h|y}(X_{\theta,y}) \otimes Z_{\theta,y,z})] \\ = \text{tr}[G_{j|yz}(X_{\theta,y} \otimes Z_{\theta,y,z})]. \end{aligned}$$

This is the adjoint identity, with no division by a history probability. It proves the assertion also at null histories. For countably many outcomes the sums of positive effects increase weakly to bounded effects, and the corresponding probabilities converge by Lemma 2.2. Equivalently retain a remainder branch and pass to its vanishing trace tail. Continuity then covers norm limits of finite testers. No new quantum system was introduced; the old system retained by the replacement is precisely R , and the fresh reference remains F_y . Compression to a common support and extension of the POVM by an arbitrary decision on its complement give the last assertion. $\square$

A constraint only on the smaller *post-instrument* system is not sufficient for this resource statement: postponing a compression can retain a larger R . We do not apply the lemma to bypass such a cut. When σ_{yh} depends on h , there need not be one common $Z_{\theta,y,z}$, so the displayed factorization is unavailable. First-label dependence σ_y causes no such difficulty.

3.2. An exact criterion for arbitrary finite experiments. Retain the experiment and the positive payoff matrices (2.3). Fix a pure first acquisition $|C_0\rangle$ with $C_0 : I_1 \rightarrow R$, $C_0^* C_0 = \rho$, and compress R to the range of C_0 . The initial state is held fixed, not replaced by an initially announced mixture. Let $1 \leq \ell \leq d_2$ and write

$$g_{y,\ell}(a) = \max_{b \in \mathcal{D}_{d_2}, \text{rank } b \leq \ell} \Phi_y(a, b), \quad c_{y,\ell}(\rho) = \max_{\rho = \sum_h w_h a_h, w_h \geq 0, a_h \in \mathcal{D}_{d_1}} \sum_h w_h g_{y,\ell}(a_h).$$

Here Φ_y is the final-POVM optimization in (2.4). The variables $w_h = \text{tr } A_h$ are *Gram weights*; the physical first-record probability is $w_h \text{tr}(a_h(E_y^{\theta,1})^\top)$. The two are not interchangeable. Both old receivers and their intermediates may be retained without a bound below d_1 . Write $P_{\text{label},\ell}(\rho)$ and $P_{\text{msg},\ell}(\rho)$ for the fixed-initial optima when fresh preparation may use only y , or may also use the receiver message.

Theorem 3.2 (Uniform message criterion). *For every finite experiment, every prescribed pure initial Gram ρ , and every fresh-reference bound ℓ ,*

$$(3.2) \quad \begin{aligned} P_{\text{label},\ell}(\rho) &= \sum_y g_{y,\ell}(\rho), \\ P_{\text{msg},\ell}(\rho) &= \sum_y c_{y,\ell}(\rho), \\ \Delta_\ell(\rho) &= \sum_y [c_{y,\ell}(\rho) - g_{y,\ell}(\rho)] \geq 0. \end{aligned}$$

Both optima are attained; for the second, at most $(\text{rank } \rho)^2$ nonzero receiver outcomes after each first label suffice. The following statements are equivalent:

- (i) $\Delta_\ell(\rho) = 0$ for every prescribed pure first acquisition ρ ;
- (ii) every $g_{y,\ell}$ is concave on $\mathcal{D}_{d_1}$;
- (iii) no protocol with a binary receiver instrument after one first label, and no receiver instrument after the other labels, can strictly outperform the optimal label-only value at its own prescribed initial acquisition.

If messages help somewhere, two outcomes after a single first label therefore suffice to witness a strict improvement at some initial acquisition. This last assertion does not fix in advance the initial acquisition at which the witness occurs.

Proof. Square roots, payoff contractions and the compact final-POVM optimization make Φ_y continuous. The set of rank-at-most- ℓ densities is compact, so $g_{y,\ell}$ is continuous and its maximum is attained. A mixed fresh preparation with reference dimension ℓ may be decomposed into pure vectors on that same space. With no receiver message, their mixture is independent of the old record. Linearity shows that one of these fresh vectors does at least as well as the mixture for a fixed continuation. Its Gram has rank at most ℓ ; conversely every such Gram has a purification with reference dimension ℓ . Lemma 3.1 now proves the first formula without disallowing old quantum information in the final readout.

With messages, resolve receiver Kraus indices and fresh pure-state mixtures into the public record. Every fresh pure coefficient map has rank at most ℓ on its existing reference; refining the receiver instrument keeps its output space unchanged. The resulting branch Grams have the common sum ρ for every y , and the best branch value is $w_h g_{y,\ell}(a_h)$. Conversely Lemma 2.3 implements every finite such decomposition on the fixed Schmidt support; the fresh factors have rank at most ℓ . The graph of $g_{y,\ell}$ on the density matrices supported in $\text{supp } \rho$ is compact. Its convex hull gives an attaining roof. If an attaining ensemble has more than s^2 atoms, $s = \text{rank } \rho$, its density matrices are linearly dependent in a real vector space of dimension s^2 . Vary their weights in both directions along this dependence. An objective-changing direction would contradict

optimality; otherwise move until a weight vanishes. Iterating proves the atom bound. Countable outcomes follow by positive tails. These arguments prove the second formula and its attainment.

Each difference in the last line of (3.2) is nonnegative. If (i) holds, each difference vanishes at every ρ ; thus $g_{y,\ell} = c_{y,\ell}$ and (ii) follows. Conversely concavity makes each roof equal to its defining function, proving (i). If (ii) fails, by the definition of concavity there are a label y_0 , states a, b , and $0 < \lambda < 1$ such that, for $\rho = \lambda a + (1 - \lambda)b$,

$$\lambda g_{y_0,\ell}(a) + (1 - \lambda)g_{y_0,\ell}(b) > g_{y_0,\ell}(\rho).$$

Realize this two-atom instrument after y_0 , with its optimizing fresh continuations. At every other first label use the optimizing label-only continuation at ρ . The strict displayed excess is exactly its excess over the *optimal* label-only value at ρ . This disproves (iii). Finally (i) implies (iii) by class inclusion. $\square$

The criterion applies to fixed-pair as well as ensemble problems, but it does not assert that a particular experiment has a positive gap. It classifies uniform absence of message advantage in the stated reset interface, not every meaning of bounded coherent memory. The inclusion diagram at a fixed admissible initial acquisition is

$$\mathfrak{C}_\ell(\rho) \subseteq \mathfrak{R}_{d_1,\ell}(\rho) \supseteq \mathfrak{L}_\ell(\rho), \\ \mathfrak{R}_{d_1,\ell}(\rho) \subseteq \mathfrak{A}(\rho).$$

where $\mathfrak{C}$ keeps only classical old information, $\mathfrak{L}$ allows label feedback and full old quantum storage but no receiver-to-source message, and $\mathfrak{A}$ allows coherent acquisition. No inclusion between $\mathfrak{C}$ and $\mathfrak{L}$ is used. At a fixed Gram, a message gap is compatible with a globally optimized no-message strategy using a different first acquisition.

Corollary 3.3 (Stability of the message gap). *Keep the labels, reward table, prior law, and both optimization classes fixed. For every latent index θ , suppose the entire slot- i measurement channel changes by at most δ_i in unhalved diamond norm, $i = 1, 2$. The two optima in (3.2) each change by at most $(\delta_1 + \delta_2)/2$, and*

$$|\tilde{\Delta}_\ell(\rho) - \Delta_\ell(\rho)| \leq \delta_1 + \delta_2.$$

In particular a larger original gap remains strict under these perturbations. The latent device is still selected once.

Proof. For a fixed protocol, replace the two channel slots in turn. Diamond-norm control, including every reference, and trace-norm contractivity bound the change of its final normalized state by $\delta_1 + \delta_2$. The difference has trace zero, so a reward effect between zero and the identity changes its expectation by at most one half of that norm. Average over the fixed prior and take the supremum in each unchanged class. The triangle inequality gives the gap bound. The argument compares specified valid reset protocols; it does not infer physical reset independence from the observed score. $\square$

4. AN EXACT OPERATIONAL SEPARATION OF ACQUISITION AND RECEIVER MEMORY

The local profile law does not identify exact operational distances. We now separate three strategy classes on one finite ensemble of ordered measurement channels. The separation holds in every input dimension, throughout a depolarizing family, with exact optimal success probabilities. The unknown interface still has only classical output. The experiment has a binary hypothesis decision, not a request to identify every member of the ensemble.

4.1. A finite family and its second moments. Let $d \geq 2$, let S interchange the two factors of $\mathbb{C}^d \otimes \mathbb{C}^d$, and put $\Pi_{\pm} = (I \pm S)/2$. A weighted family of ordered orthonormal bases consists of projections P_y^b , $1 \leq y \leq d$, and positive weights w_b summing to one. The weights are real; rational or algebraic coordinates are not required by the theorem. We call this a weighted exact second-moment family, not necessarily an unweighted projective design. The required identities are

$$(4.1) \quad \sum_b w_b P_y^b \otimes P_z^b = \begin{cases} (I + S)/(d(d+1)), & y = z, \\ (dI - S)/(d(d^2 - 1)), & y \neq z. \end{cases}$$

Lemma 4.1 (Finite simultaneous second moments). *In every dimension $d \geq 2$ there is such a family with at most $d^6 + 1$ bases. For $d = 2$, the six ordered eigenbases of the three Pauli matrices, with both orders and equal weights, satisfy (4.1).*

Proof. Average the ordered coordinate basis over Haar measure on $U(d)$. The second moment of a single column is supported on the symmetric subspace, invariant under $U \otimes U$, and has trace one. It is therefore $2\Pi_+/[d(d+1)]$. For distinct columns the invariant moment has the form $aI + bS$; its trace is one and its contraction with S is zero, since the columns are orthogonal. These two equations give the second line of (4.1). Equivalently, this line follows by summing over the other $d-1$ columns and using the first moment I/d . The vector of all d^2 Hermitian second moments lies in a real space of dimension at most d^6 . Its Haar average belongs to the convex hull of the compact set of these vectors. Carathéodory's theorem gives a convex combination of at most $d^6 + 1$ vectors; discard zero weights. For the stated qubit family expand $P = (I \pm \sigma_a)/2$ and use $S = (I \otimes I + \sum_{a=1}^3 \sigma_a \otimes \sigma_a)/2$. $\square$

This is a finite existence argument, not an efficient design construction. Second-moment designs themselves are established objects [5].

Fix $0 \leq t \leq 1$, and consider the ordered measurements

$$(4.2) \quad C_y = I/d, \quad E_y^b(t) = (1-t)I/d + tP_y^b.$$

Set

$$(4.3) \quad L = 2(d+1) - t^2, \quad p_C = \frac{d+1-t^2}{L}, \quad p_E = \frac{d+1}{L}.$$

With probability p_C the device is C ; otherwise a basis b is chosen with law w_b and the device is $E^b(t)$. This choice is made once. Both calls use the same chosen, memoryless, classical-output device. The task is to decide C versus the alternative family. In particular, the second moments cannot be replaced by two independent first moments. Every effect is strictly positive for $t < 1$. Taking the partial trace in (4.1) gives $\sum_b w_b E_y^b(t) = C_y$. Thus the two hypotheses have identical one-call channels, including on entangled inputs, and the one-call optimum is p_E . The two-call signal concerns a device chosen once; it is absent if the alternative is independently redrawn at each call.

We compare: (i) complete first-call measurement followed by measure-and-reprepare classical adaptation, as in [2, Definition 23]; (ii) unit-reset protocols of Definition S12.1, with profile $(1, 1)$ and arbitrary receiver quantum memory and classical feedback; and (iii) all two-call adaptive protocols. Let their optimal Bayes success probabilities be P_{ca} , P_{reset} and P_{all} , respectively. Early stopping is allowed and can be implemented by a discarded fresh call. The three closed optimization classes are precisely those of Definition 2.1. Lemma 2.3 gives the full branch normal form, and Proposition 2.10 gives the physical causal normalization independently of this example.

Theorem 4.2 (Exact two-call hierarchy). *The alternative basis index in this experiment is drawn once and reused at both calls. Independently redrawing it instead gives the scalar product channel*

and no improvement over the larger prior. For every finite family satisfying (4.1), every $d \geq 2$, and every $0 \leq t \leq 1$, the game (4.2)–(4.3) has the exact optimal values

$$(4.4) \quad \begin{aligned} P_{\text{ca}} &= \frac{d+1}{L}, \\ P_{\text{reset}} &= \frac{d+1}{L} + \frac{t^2(d-1)}{2dL}, \\ P_{\text{all}} &= \frac{d+1+t^2}{L}. \end{aligned}$$

For $t > 0$ both inequalities are strict. At $t = 0$ all three values are $1/2$. The priors sum to one and $p_E - p_C = t^2/L$, so p_E is the constant-decision baseline. The middle value is attained by independent maximally entangled input–reference pairs and a final joint reference readout, without feedback. Its upper bound allows all unit-reset feedback and unrestricted receiver dimension. The last value is attained by a parallel antisymmetric two-input state, and its upper bound allows every adaptive two-call tester.

The theorem separates a finite ensemble decision task, not an arbitrary pair of fixed channels and not a complete memory-dimension hierarchy. The middle class has hard resources $N = Q = 2$; the last construction is one acquisition group of size two, with square cost four.

4.2. A swap-filter identity. For a Hermitian matrix X , write X_+ for its positive part.

Lemma 4.3 (Negative mass of a filtered swap). *Let $A, B \succeq 0$ be $d \times d$ matrices and $K = (\sqrt{A} \otimes \sqrt{B})S(\sqrt{A} \otimes \sqrt{B})$. Then*

$$(4.5) \quad \text{tr}(-K)_+ = \frac{1}{2} \left[(\text{tr} \sqrt{\sqrt{A}B\sqrt{A}})^2 - \text{tr}(AB) \right] \leq \frac{d-1}{2d} \text{tr} A \text{tr} B.$$

The same negative mass is obtained from $(U \otimes V)S(U \otimes V)^*$ when $U^*U = A$ and $V^*V = B$, even if the receiving spaces are larger. If $\text{tr} A \text{tr} B > 0$, equality in the upper bound holds if and only if $A = (\text{tr} A)I/d$ and $B = (\text{tr} B)I/d$.

Proof. First suppose A, B are positive definite. The matrix K is similar to $(A \otimes B)S$. Conjugation of the latter by $I \otimes B$ gives $(AB \otimes I)S$. The matrix AB is similar to $\sqrt{A}B\sqrt{A}$; let its positive eigenvalues be $\lambda_1, \dots, \lambda_d$. Conjugating by the tensor square of an eigenbasis matrix leaves S unchanged. The resulting matrix has eigenvalues λ_i on the one-dimensional spaces indexed by (i, i) , and $\pm\sqrt{\lambda_i\lambda_j}$ on the two-dimensional spaces indexed by $(i, j), (j, i)$, $i < j$. Thus its negative mass is $\sum_{i < j} \sqrt{\lambda_i\lambda_j}$, which gives the equality. By Cauchy–Schwarz, $\sum_i \lambda_i \geq d^{-1}(\sum_i \sqrt{\lambda_i})^2$. Moreover, the polar decomposition and Hilbert–Schmidt Cauchy–Schwarz give $\|\sqrt{A}\sqrt{B}\|_1 \leq \sqrt{\text{tr} A \text{tr} B}$. These inequalities give the bound. Approximation by positive definite matrices proves the singular case. The polar decompositions of U, V identify their filtered swap with K on its support, adding only zero eigenvalues in the orthogonal complement. For the equality assertion, put $f = \|\sqrt{A}\sqrt{B}\|_1$ and $a = \text{tr} A$, $b = \text{tr} B$. Equality in the stated bound, with $ab > 0$, forces equality in both $\text{tr}(AB) \geq f^2/d$ and $f^2 \leq ab$. The variational formula for the trace norm and equality in Hilbert–Schmidt Cauchy–Schwarz imply $B = (b/a)A$: indeed a maximizing unitary U has $\sqrt{B}U = c\sqrt{A}$, and multiplication by its adjoint gives this proportionality. Equality in the first inequality makes all eigenvalues of $\sqrt{A}B\sqrt{A}$ equal and positive. Consequently $A = (a/d)I$ and $B = (b/d)I$. These scalar matrices plainly attain the bound. This also excludes equality at nonzero singular factors. $\square$

The normalized equality statement and its singular case are expanded in the proof of Lemma 5.1: normalized fidelity one forces equality of the two density matrices, and eigenvalue Cauchy–Schwarz then forces all d eigenvalues to be $1/d$.

4.3. Payoff and complete normalization. We use the fully transposed, unnormalized Choi representative

$$(4.6) \quad J^\sharp(E) = [(\text{Id} \otimes \mathcal{M}_E)(|\Omega\rangle\langle\Omega|)]^\top = \sum_y E_y \otimes |y\rangle\langle y|, \quad |\Omega\rangle = \sum_i |ii\rangle.$$

Its output trace is I . A simultaneous full transpose of the usual tester representative makes its contraction with $J^\sharp$ the Born probability. Factor order is I_1, Y_1, I_2, Y_2 ; expressions grouping I_1, I_2 use the canonical permutation of these factors. Dephasing the output factors does not change any score. Write $T_{0,yz}$ for the input block of the tester decision C . The gain over always guessing the alternative is

$$(4.7) \quad P - p_E = \sum_{y,z} \text{tr}(T_{0,yz} W_{yz}), \quad W_{yz} = p_C C_y \otimes C_z - p_E \sum_b w_b E_y^b(t) \otimes E_z^b(t).$$

The averaged tensor in the second term is $(1 - t^2)I/d^2$ plus t^2 times (4.1). With $k = t^2/(dL)$, substitution gives

$$(4.8) \quad W_{yy} = -kS, \quad W_{yz} = -\frac{2k}{d-1}\Pi_- \quad (y \neq z).$$

These operators are real in the coordinate basis; a full transpose therefore does not alter the following congruences.

Proof of Theorem 4.2. Classical adaptation. Each block of a measure-and-reprepare tester is a sum of positive products $A \otimes B$. Since $\text{tr}[(A \otimes B)S] = \text{tr}(AB) \geq 0$, equal labels contribute no positive gain in (4.7); different labels cannot contribute positively either. Hence $P_{\text{ca}} \leq p_E$, with equality by the constant decision. This proof also covers limits of such testers.

All unit-reset protocols. Purify the first acquisition and write its state as $\sum_i |i\rangle C_0 |i\rangle$, with $\text{tr} C_0^* C_0 = 1$. After the first device label y , combine all intervening receiver operations into an instrument with recorded outcome h . For an upper bound, retain its Stinespring environment in the receiver. The first conditional reference operator then has the form $C_{yh} E_y^\top C_{yh}^*$, where

$$A_{yh} = C_{yh}^* C_{yh}, \quad \sum_h A_{yh} = C_0^* C_0 \quad \text{for each } y.$$

Retaining this environment only enlarges receiver processing and does not import it into the second acquisition. Conditional reset permits a purification of that fresh acquisition with a map D_{yh} satisfying $B_{yh} = D_{yh}^* D_{yh}$ and $\text{tr} B_{yh} = 1$. Its complete system is independent of the old receiver conditional on (y, h) . The two-reference payoff operator at record (y, h, z) is therefore

$$(C_{yh} \otimes D_{yh}) W_{yz}^\top (C_{yh} \otimes D_{yh})^*.$$

For different labels it is nonpositive. For equal labels, maximizing any final effect bounded by the identity gives at most its positive mass, which by Lemma 4.3 is $k(d-1) \text{tr} A_{yh}/(2d)$. Summing over y, h gives

$$P_{\text{reset}} - p_E \leq k(d-1)/2 = \frac{t^2(d-1)}{2dL}.$$

Public randomization can be included in the recorded history. A normal instrument with countably many outcomes is treated by trace-class sums, or by finite truncations retaining a remainder outcome and then taking the trace-norm limit. The bound is independent of receiver size. Thus no finite-dimensional restriction on receiver storage has entered the upper bound. For $t > 0$, the equality case of Lemma 4.3 also shows that a nonrandomized protocol attaining this bound in the displayed purified normal form must satisfy $A_{yh} = (\text{tr} A_{yh})I/d$ and $B_{yh} = I/d$ on every nonzero branch. All deficits in the summed upper bound are nonnegative, so equality forces their branchwise vanishing. Summing over h then gives $C_0^* C_0 = I/d$. Maximal mixing is thus a

consequence of equality, not a restriction imposed on the competing protocols. This statement concerns the purified normal form, not uniqueness of its dilations or of the final readout.

For attainment, prepare $|\Omega\rangle/\sqrt{d}$ independently on each input and reference. Declare C precisely when the two device labels agree and the references lie in Π_- . The event probability under C is $a = (d-1)/(2d^2)$. Under any $E^b(t)$ it is $(1-t^2)a$: indeed $\text{tr } E_y^b(t) = 1$, $\text{tr } E_y^b(t)^2 = t^2 + (1-t^2)/d$, and $\text{tr}[\Pi_-(E \otimes E)] = [(\text{tr } E)^2 - \text{tr } E^2]/2$. The factor d^{-2} comes from the two normalized input-reference pairs. The resulting gain is $a[p_C - p_E(1-t^2)] = t^2(d-1)/(2dL)$.

All adaptive protocols. The deterministic normalization of any two-call tester is $T_0 + T_1 = \Xi \otimes I_{Y_2}$, where, after dephasing Y_1 , Ξ has input blocks $\Xi_y \succeq 0$ satisfying

$$\text{tr}_{I_2} \Xi_y = \rho \quad (1 \leq y \leq d), \quad \text{tr } \rho = 1.$$

Here is a direct realization of these constraints in the present classical-output interface. Purify the initial state as $|\psi\rangle \in I_1 \otimes R$, and let $\mathcal{A}_y$ be the trace-preserving intervening map from R to $I_2 \otimes R'$ after observing y . Put

$$X_y = (\text{Id}_{I_1} \otimes \mathcal{A}_y)(|\psi\rangle\langle\psi|), \quad \Xi_y = \text{tr}_{R'} X_y, \quad \rho = \text{tr}_R |\psi\rangle\langle\psi|.$$

Positivity and trace preservation give the displayed marginal equations. If F_{yz} is the final receiver effect for decision C , then

$$T_{0,yz} = \text{tr}_{R'} [(I_{I_1 I_2} \otimes \sqrt{F_{yz}}) X_y (I_{I_1 I_2} \otimes \sqrt{F_{yz}})].$$

The Choi contraction (4.6) gives its Born probability, and $0 \preceq F_{yz} \preceq I$ gives $0 \preceq T_{0,yz} \preceq \Xi_y$. This also proves positivity of the complement, without an input-dimension factor from a normalized Choi state. In particular $\text{tr } \Xi_y = 1$. Since $-kS \preceq kI$ and the other payoff blocks are nonpositive,

$$P_{\text{all}} - p_E \leq k \sum_y \text{tr } \Xi_y = kd = t^2/L.$$

Prepare any normalized antisymmetric pure state on the two inputs and declare C when their labels agree. The event probability is $1/d$ under C , and $(1-t^2)/d$ under every $E^b(t)$, because $(P_y^b \otimes P_y^b)\Pi_- = 0$. The gain is t^2/L . This is a legitimate parallel two-call strategy and attains the adaptive upper. The case $t = 0$ gives equal values $1/2$ directly. $\square$

Corollary 4.4 (Seven qubit devices). *Take the scalar qubit measurement with prior $2/5$ and each of the six ordered Pauli measurements with prior $1/10$. With the device fixed for both calls and the binary decision specified above, the exact success probabilities are*

$$P_{\text{ca}} = 3/5, \quad P_{\text{reset}} = 13/20, \quad P_{\text{all}} = 4/5.$$

Proof. Use Lemma 4.1 and substitute $d = 2, t = 1$. $\square$

The antisymmetric readout and comparison mechanism have antecedents in measurement comparison [6]; reductions between measurement and state discrimination also have a substantial history [1]. Here the assertion is the joint exact optimization over the three specified classes, including unrestricted reset feedback, on one finite classical-output ensemble and its full-rank deformation.

4.4. Quantitative stability of the operational gap.

Corollary 4.5 (A stable score separation). *Suppose every device in the finite game is replaced by a classical-output measurement channel at diamond distance at most δ from its specified channel. Keep the priors and the two-call resource classes fixed. Each of their optimal success probabilities changes by at most δ . If the displayed reset implementation also has total hard*

pathwise preparation defect at most ε in the sense of Proposition S14.4, its score exceeds the optimal perturbed classical-adaptation score whenever

$$(4.9) \quad 2\delta + \varepsilon/2 < \frac{t^2(d-1)}{2dL}.$$

Proof. A two-call hybrid changes each normalized output in unhalved trace norm by at most 2δ . A fixed decision event changes by at most δ , and averaging over the unchanged priors preserves this bound. Taking suprema over a fixed strategy class gives the same two-sided bound for its optimum. The preparation certificate changes each implemented output by at most ε , hence its decision probability by at most $\varepsilon/2$. Subtract the classical upper from the reset lower in (4.4). $\square$

These are conditional norm estimates, not physical calibration claims. In the preceding corollary the same δ bounds the scalar channel and each entire basis measurement channel individually. Priors and the once-drawn-index law stay fixed; index-dependent channel perturbations are permitted. Corollary 5.3 additionally treats prior and law perturbations, with the unhalved norm factors explicit. They make the relative scale needed to preserve a small score gap explicit.

4.5. Hard budget conventions beyond the normalized domain.

Proposition 4.6 (Normalization of arbitrary hard thresholds). *For positive integer reservations, $N_\pi \leq Q_\pi \leq N_\pi^2$. Given real hard bounds $n, q \geq 1$, put*

$$M = \min\{\lfloor n \rfloor, \lfloor q \rfloor\}, \quad R = \min\{\lfloor q \rfloor, M^2\}.$$

The class $\{\pi : N_\pi \leq n, Q_\pi \leq q\}$ equals $\{\pi : N_\pi \leq M, Q_\pi \leq R\}$, and $M \leq R \leq M^2$. Consequently Theorem S14.3 applies with (M, R) .

Proof. On each path $\sum n(h) \leq \sum n(h)^2 \leq (\sum n(h))^2$. Taking hard suprema proves the first assertion. Bounded integer costs have integer suprema, so both constraints can be floored. The first inequality reduces the call bound to M and the last reduces the quadratic bound to R . Conversely the reduced bounds imply the original ones. Finally $M \leq \lfloor q \rfloor$ and $M \leq M^2$. $\square$

The normalization $N \leq Q$ allows the N -singleton lower construction; it does not say that every admissible individual policy has $Q_\pi \geq N$. If a hard threshold is below one, only the zero-call class is available. Fractional-duration reservations and expected-cost constraints are different models. None is substituted for a hard integer reservation here.

5. AN EXACT HIERARCHY WITH EQUAL HYPOTHESIS PRIORS

The strict separation does not require the prior in (4.3). Equal hypothesis priors lead to a different optimal receiver readout and to another exact hierarchy. The finite weighted second-moment family is unchanged.

5.1. A trace-norm bound. Let S be the swap on $\mathbb{C}^d \otimes \mathbb{C}^d$ and $\Pi_\pm = (I \pm S)/2$.

Lemma 5.1 (The centered filtered swap). *For $d \geq 2$, $A, B \succeq 0$, and $L = \sqrt{A} \otimes \sqrt{B}$,*

$$(5.1) \quad \|L(I - dS)L\|_1 \leq \left(d - \frac{1}{d}\right) \operatorname{tr} A \operatorname{tr} B.$$

For nonzero factors equality holds precisely when A and B are positive scalar multiples of I . The same norm and bound hold for rectangular factors having Gram matrices A, B .

Proof. Put $K = LSL$ and $f = \|\sqrt{A}\sqrt{B}\|_1$. The spectrum calculation in Lemma 4.3 gives $\|K\|_1 = f^2$, including singular factors by continuity. Since $I - dS = 2\Pi_- - (d-1)S$, the triangle inequality gives

$$\begin{aligned} \|L(I - dS)L\|_1 &\leq 2\operatorname{tr}(L\Pi_-L) + (d-1)\|K\|_1 \\ &= \operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB) + (d-1)f^2 \\ &\leq \operatorname{tr} A \operatorname{tr} B + (d-1-d^{-1})f^2 \\ &\leq (d-d^{-1})\operatorname{tr} A \operatorname{tr} B. \end{aligned}$$

The penultimate step is eigenvalue Cauchy–Schwarz, $\operatorname{tr}(AB) \geq f^2/d$; the last is $f^2 \leq \operatorname{tr} A \operatorname{tr} B$.

Here is the equality argument, also supplying a detailed treatment of singular nonzero factors in Lemma 4.3. Normalize $a = A/\operatorname{tr} A$, $b = B/\operatorname{tr} B$. The variational formula for the trace norm selects a unitary U with $\operatorname{Re} \operatorname{tr}(\sqrt{a}\sqrt{b}U) = \|\sqrt{a}\sqrt{b}\|_1$. Both $\sqrt{a}$ and $\sqrt{b}U$ have Hilbert–Schmidt norm one. Equality in Cauchy–Schwarz, with the phase chosen as above, forces $\sqrt{b}U = \sqrt{a}$; multiplication by the adjoint gives $b = a$. This implication uses no invertibility. Equality in $\operatorname{tr}(AB) \geq f^2/d$ then gives $\operatorname{tr}(a^2) = 1/d$. The d nonnegative eigenvalues of a , including zeros, sum to one, so equality in their Cauchy–Schwarz inequality forces all of them to equal $1/d$. Thus nonzero singular factors cannot attain equality. Since $d-1-d^{-1} > 0$, equality in (5.1) forces both of these equalities. Scalar factors attain the bound directly from the eigenvalues $1-d$ on Π_+ and $1+d$ on Π_- . Polar decompositions of rectangular factors preserve the nonzero spectrum and add only zero eigenvalues, proving the last assertion. $\square$

5.2. Equal-prior optimization.

Theorem 5.2 (Equal-prior operational hierarchy). *Let (w_b, P_y^b) be any weighted exact second-moment family satisfying (4.1). Choose the scalar measurement $C_y = I/d$ with probability $1/2$, or choose the alternative family with probability $1/2$ and then draw one basis index b with law w_b . In the latter case the same device $E_y^b(t) = (1-t)I/d + tP_y^b$ is used at both calls. For $d \geq 2$ and $0 \leq t \leq 1$, the three classes of Definition 2.1 have exact optimal success probabilities*

$$\begin{aligned} P_{\text{ca}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(d-1)}{2d(d+1)}, \\ P_{\text{reset}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(d-1)}{2d^2}, \\ P_{\text{all}}^{\text{eq}} &= \frac{1}{2} + \frac{t^2}{2d}. \end{aligned} \tag{5.2}$$

Both inequalities are strict for $t > 0$, including the full-rank interval $0 < t < 1$; all three values equal $1/2$ at $t = 0$. The reset optimum is attained with independent maximally entangled input–reference pairs and no receiver feedback. Its upper includes all unit-reset receiver instruments and arbitrary receiver dimension. The unrestricted optimum is attained by a normalized antisymmetric two-input state and bounds every adaptive strategy. If the alternative basis is instead redrawn independently at each call, its two-slot averaged channel is the scalar product channel and all three equal-prior values are $1/2$.

Proof. Put $a = t^2/[2d^2(d+1)]$. Subtracting the alternative-weighted second moment from the scalar-weighted product gives

$$W_{yy} = a(I - dS), \quad W_{yz} = -\frac{a}{d-1}(I - dS) \quad (y \neq z). \tag{5.3}$$

Indeed the diagonal moment is $(1-t^2)I/d^2 + t^2(I+S)/[d(d+1)]$, and the off-diagonal moment is $(1-t^2)I/d^2 + t^2(dI-S)/[d(d^2-1)]$. These substitutions prove (5.3). Both S and $\Pi_{\pm}$ are real in

the chosen basis; their full transpose does not alter these payoff blocks. The gain over the constant alternative decision is $\sum_{yz} \text{tr}(T_{C,yz} W_{yz})$.

Classically adaptive upper. After a complete first readout, a branch has an input effect $A \succeq 0$, with $\sum_h A_{yh} = \rho_1$ for each y and $\text{tr} \rho_1 = 1$. The second single-call tester has normalization $B \succeq 0$, $\text{tr} B = 1$, and decision blocks $0 \preceq T_z \preceq B$. For $\text{tr} A > 0$ set $\hat{A} = A / \text{tr} A$. Contracting (5.3) with A in the first slot gives $a \text{tr} A(I - d\hat{A})$ for $z = y$ and its negative divided by $d - 1$ for $z \neq y$. For a Hermitian Q and $0 \preceq T \preceq B$, $\text{tr}(TQ) \leq \text{tr}(BQ_+)$. Therefore the branch gain is at most

$$a \text{tr} A \text{tr}[B|I - d\hat{A}|] \leq a(d - 1) \text{tr} A.$$

The final inequality follows from $0 \preceq \hat{A} \preceq I$ and $d \geq 2$. The zero branch contributes zero. Summing $\sum_{yh} \text{tr} A_{yh} = d$ proves the first upper bound. It covers arbitrary second-call references and decisions, not only unassisted classical probes. Norm closure preserves it.

To attain it, use the same fixed pure input $|u\rangle$ twice and declare C on unequal labels. Under C the equal-label probability is $1/d$; under the averaged alternative it is

$$\sum_{by} w_b \langle u, E_y^b(t) u \rangle^2 = \frac{1}{d} + \frac{t^2(d - 1)}{d(d + 1)}.$$

This follows directly from the second moments. The corresponding Bayes gain is half their difference, namely $ad(d - 1)$.

Unit-reset upper. Use Lemma 2.3 without any restriction on its rectangular receiver spaces. On a branch (y, h) put

$$X = (C_{yh} \otimes D_{yh})(I - dS)(C_{yh} \otimes D_{yh})^*.$$

The elementary identity $\max_{0 \preceq F \preceq I} \text{tr}(FX) = \text{tr}(X_+)$ and (5.3) give total branch gain at most

$$a \text{tr}(X_+) + a \text{tr}((-X)_+) = a\|X\|_1 \leq a(d - d^{-1}) \text{tr} A_{yh},$$

since $\text{tr} b_{yh} = 1$. There are $d - 1$ opposite-sign output blocks, which cancel their denominator. Summing over y, h gives $a(d^2 - 1) = t^2(d - 1)/(2d^2)$. Countable outcomes and null histories use unnormalized positive tails.

For attainment, prepare two independent normalized maximally entangled pairs $|\Omega\rangle/\sqrt{d}$. On equal device labels declare C when the references lie in Π_- ; on unequal labels declare C when the references lie in Π_+ . This is one final joint receiver measurement, with complementary effect $I - F_C$ on each classical block. Its event probabilities are

$$(5.4) \quad q_C = \frac{(d - 1)(d + 2)}{2d^2}, \quad q_b = \frac{(d - 1)(d + 2 - 2t^2)}{2d^2} \quad \text{for every } b.$$

To verify them, the receiver block is $(E_y^b(t))^\top \otimes (E_z^b(t))^\top / d^2$ and $\text{tr}[\Pi_\pm(A \otimes B)] = [\text{tr} A \text{tr} B \pm \text{tr}(AB)]/2$. Here $\text{tr} E_y = 1$, $\text{tr} E_y^2 = t^2 + (1 - t^2)/d$, and $\text{tr} E_y E_z = (1 - t^2)/d$ for $y \neq z$. Counting the d equal-label and $d(d - 1)$ unequal-label blocks gives (5.4). Half the difference is the reset upper.

Unrestricted upper. By Proposition 2.10, $T_{C,yz} \preceq \Xi_y$ and $\text{tr} \Xi_y = 1$. The sum of the positive payoff parts for a fixed first label is

$$(W_{yy})_+ + \sum_{z \neq y} (W_{yz})_+ = a(d + 1)\Pi_- + a(d - 1)\Pi_+ \preceq a(d + 1)I.$$

Thus the gain is at most $ad(d + 1) = t^2/(2d)$. Any normalized state supported on the nonzero antisymmetric input subspace attains it by declaring C on equal labels: their probability is $1/d$ under C and $(1 - t^2)/d$ under every alternative.

The gaps are

$$(5.5) \quad P_{\text{reset}}^{\text{eq}} - P_{\text{ca}}^{\text{eq}} = \frac{t^2(d-1)}{2d^2(d+1)}, \quad P_{\text{all}}^{\text{eq}} - P_{\text{reset}}^{\text{eq}} = \frac{t^2}{2d^2}.$$

Finally $\sum_b w_b \mathcal{M}_{E^b(t)} = \mathcal{M}_C$ as a channel identity. Two independent draws give its tensor square, hence the same two-slot statistics even for adaptive testers. This proves the last assertion and the endpoint statements. $\square$

The equal priors concern the two hypotheses, not a uniform prior on all individual devices. The result removes prior tuning, but still uses a shared latent device. It does not assert a gap for every fixed pair. Together with Theorem 2.4, it gives a general optimization principle and an explicitly solved family, rather than identifying these two levels of generality.

5.3. Seven explicit qubit devices. For $d = 2$ write

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The devices are $C = (I/2, I/2)$ and, for $A \in \{X, Y, Z\}$ and $\epsilon \in \{-1, 1\}$,

$$E^{A,\epsilon}(t) = \left(\frac{I + t\epsilon A}{2}, \frac{I - t\epsilon A}{2} \right).$$

The six alternatives have conditional weight $1/6$. Both outcome orders are included: they cancel the first Pauli moments and supply the required ordered second moments. At $t = 1$ they are the six ordered Pauli eigenbasis measurements. For the prior of Theorem 4.2, C has mass $2/5$ and each alternative mass $1/10$; for Theorem 5.2, these masses are $1/2$ and $1/12$.

For reset acquisition prepare

$$|\phi^+\rangle_{I_1 R_1} \otimes |\phi^+\rangle_{I_2 R_2}, \quad |\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}.$$

The inputs are independent. The two retained references are measured jointly only after both calls. Let $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$, $\Pi_- = |\psi^-\rangle\langle\psi^-|$ and $\Pi_+ = I_4 - \Pi_-$ on $R_1 \otimes R_2$. The receiver block for any two devices is $E_y^\top \otimes F_z^\top/4$. Under $J^\sharp$, the decision tester block is $F_{C,yz}^\top/4$, where

$$F_{C,yz} = \mathbf{1}_{y=z} \Pi_- \quad \text{for the prior } 2/5, \quad F_{C,yz} = \begin{cases} \Pi_-, & y = z, \\ \Pi_+, & y \neq z \end{cases} \quad \text{for equal priors.}$$

The other block is $(I_4 - F_{C,yz})^\top/4 \succeq 0$. Their sum is $I_4/4$ for each (y, z) , with partial trace $I_2/2$ over I_2 and trace one. This verifies the causal normalization in the order I_1, Y_1, I_2, Y_2 , using the canonical permutation to group the input copies. The transpose is required for arbitrary complex receiver effects, even though the displayed antisymmetric projectors are real.

At $t = 1$, the first reset event has probabilities $1/8$ under C and zero under every alternative. Its Bayes score is $(2/5)(1/8) + (3/5) = 13/20$. The equal-prior reset event has probabilities $1/2$ and $1/4$, respectively, giving $\frac{1}{2}(\frac{1}{2}) + \frac{1}{2}(\frac{3}{4}) = 5/8$. For unrestricted acquisition put the normalized state $|\psi^-\rangle$ on $I_1 \otimes I_2$, not on the references, and declare C on equal labels. The probabilities are $1/2$ and zero, yielding $4/5$ for the first prior and $3/4$ for equal priors. For classical adaptation, the biased-prior optimum is the constant decision $3/5$. For equal priors, two inputs $|0\rangle$ and the unequal-label event have probabilities $1/2$ under C and $1/3$ averaged over alternatives, giving $7/12$. Consequently the two exact triples on the same devices are

$$\frac{3}{5} < \frac{13}{20} < \frac{4}{5}, \quad \frac{7}{12} < \frac{5}{8} < \frac{3}{4}.$$

The class upper bounds, not these evaluations alone, prove optimality.

5.4. An open neighborhood of the equal-prior experiment.

Corollary 5.3 (Channel, prior, and ensemble stability). *Fix the family and $t > 0$. Replace the scalar device and each entire basis measurement channel individually by channels within unhalved diamond distance δ . The perturbations may depend on the basis index and may change all its effects, but must form legal channels on the same interfaces. Keep the once-drawn-index rule. Replace the scalar prior $1/2$ by q and the conditional basis law w by ν , and put*

$$\eta = \frac{1}{2} \sum_b |\nu_b - w_b|, \quad \kappa = \delta + |q - \frac{1}{2}| + \eta.$$

For each of the three fixed operational classes, $|\tilde{P}_{\text{class}} - P_{\text{class}}^{\text{eq}}| \leq \kappa$. Thus both class gaps remain strict when $2\kappa < t^2(d-1)/[2d^2(d+1)]$. If the exhibited reset protocol has, in addition, a uniform reference-complete diamond certificate of preparation defect whose hard pathwise sum is at most ε , its implemented score still exceeds the perturbed classical optimum provided

$$2\kappa + \varepsilon/2 < \frac{t^2(d-1)}{2d^2(d+1)}.$$

Proof. For a fixed protocol and device index, the two-call hybrid bounds output trace distance by 2δ . Normalization makes the change of any decision probability at most δ , with the factor $1/2$ for unhalved trace distance. Changing the binary prior changes success by at most $|q - 1/2|$; changing the alternative law changes an average of $[0, 1]$ -valued successes by at most η . The bound is uniform in the protocol, so taking suprema preserves it in both directions. Subtract and use (5.5). The preparation hybrid changes each implemented output by at most ε , hence its success by at most $\varepsilon/2$. This last statement concerns a certified implementation of the displayed reset protocol, not an enlargement of the ideal optimization class. $\square$

5.5. Quantitative rigidity of near-optimal reset strategies. The equality condition for the centered swap has a stable form. It controls the input Gram matrices in a chosen purified normal form, rather than specifying unique instruments, dilations, or receiver readouts. For $d \geq 2$ set

$$c_d = d - 1 - d^{-1} > 0, \quad K_d = d \left(1 + \frac{9}{2c_d} \right).$$

Lemma 5.4 (Stable centered-swap equality). *For density matrices a, b on $\mathbb{C}^d$, let*

$$\Delta(a, b) = d - d^{-1} - \|(\sqrt{a} \otimes \sqrt{b})(I - dS)(\sqrt{a} \otimes \sqrt{b})\|_1.$$

Then $\Delta(a, b) \geq 0$ and

$$(5.6) \quad \|a - I/d\|_1^2 \leq K_d \Delta(a, b), \quad \|b - I/d\|_1^2 \leq K_d \Delta(a, b).$$

No invertibility is required.

Proof. Put $f = \|\sqrt{a}\sqrt{b}\|_1$, $\eta = 1 - f^2$, and $e = \text{tr}(ab) - f^2/d$. The proof of Lemma 5.1 gives $\eta, e \geq 0$ and

$$(5.7) \quad \Delta(a, b) \geq c_d \eta + e.$$

Choose a unitary V extending the right polar factor so that $Z = \sqrt{a}\sqrt{b}V = \sqrt{\sqrt{a}b\sqrt{a}} \succeq 0$. This is possible also when the matrices are singular. The Hilbert–Schmidt norm, denoted by $\|\cdot\|_2$, satisfies

$$\|\sqrt{b}V - \sqrt{a}\|_2^2 = 2(1 - f), \quad \|Z - a\|_2 \leq \sqrt{2\eta}, \quad \|Z - fI/d\|_2^2 = e.$$

Here $\|\sqrt{a}\|_{\text{op}} \leq 1$ and $1 - f \leq \eta$. Since also $1 - f \leq \sqrt{\eta}$, the triangle inequality yields

$$\|a - I/d\|_2 \leq (\sqrt{2} + d^{-1/2})\sqrt{\eta} + \sqrt{e}.$$

Weighted Cauchy–Schwarz bounds its square by $[1 + (\sqrt{2} + d^{-1/2})^2/c_d](c_d\eta + e)$. For $d \geq 2$, $(\sqrt{2} + d^{-1/2})^2 \leq 9/2$; and $\|T\|_1^2 \leq d\|T\|_2^2$ for a $d \times d$ matrix T . These facts and (5.7) prove the first estimate. Interchanging a, b proves the second, since swap conjugation leaves the centered-swap norm unchanged. $\square$

Corollary 5.5 (Near-optimal acquisition marginals). *Fix the equal-prior experiment of Theorem 5.2 with $t > 0$. Suppose a unit-reset strategy achieves reward $v \geq P_{\text{reset}}^{\text{eq}} - \varepsilon$. Choose its purified normal form from Lemma 2.3. On nonzero branches put*

$$a_{yh} = A_{yh}/\text{tr } A_{yh}, \quad \mu_{yh} = \text{tr } A_{yh}/d.$$

Then $\sum_{yh} \mu_{yh} = 1$ and

$$(5.8) \quad \sum_{y,h} \mu_{yh} \left(\|a_{yh} - I/d\|_1^2 + \|b_{yh} - I/d\|_1^2 \right) \leq \frac{4K_d d(d+1)}{t^2} \varepsilon.$$

For a public mixture the same conclusion holds after adjoining the seed and multiplying these weights by its law. Finite and countable recorded instruments are both allowed.

Proof. Write $\alpha = t^2/[2d^2(d+1)]$ for the coefficient in (5.3). The complete reset upper, before applying the centered-swap bound, gives

$$v \leq \frac{1}{2} + \alpha \sum_{yh} \text{tr } A_{yh} [d - d^{-1} - \Delta(a_{yh}, b_{yh})].$$

Purification does not obstruct recovering the original decision rule, so this inequality bounds its achieved reward as well. The common Gram identity gives $\sum_{yh} \text{tr } A_{yh} = d$. Subtract from the attained optimum to obtain $\sum_{yh} \mu_{yh} \Delta(a_{yh}, b_{yh}) \leq \varepsilon/(\alpha d)$. Apply Lemma 5.4 to both factors and substitute α . Positive trace tails justify countable sums. For a public mixture, apply the same upper to each component and average before subtracting its reward from the common optimum. $\square$

In particular, the total μ -weight on which either normalized Gram matrix is at trace distance at least $u > 0$ from I/d is at most $4K_d d(d+1)\varepsilon/(t^2 u^2)$. These are Gram weights, not probabilities under a selected hypothesis. At zero loss the result recovers branchwise isotropy. The estimate is nontrivial at the relative scale ε/t^2 ; it does not turn fixed calibration error into vanishing-signal robustness. It asserts neither uniqueness of a dilation nor a device-independent certification of the reset cut.

6. RANK-RESOLVED RECEIVER RESOURCES

The distinction between classical and quantum receivers extends to an attained hierarchy at each pair of retained-register dimensions. This section keeps the reset cut fixed and resolves its two sides separately. It supplies a general variational characterization and an exactly solved spectrum, rather than identifying all notions of bounded quantum memory.

6.1. Two specified recording cuts. Use the finite experiment of Definition 2.1. Fix integers $1 \leq k \leq d_1$ and $1 \leq \ell \leq d_2$.

Definition 6.1 (Retained-register dimensions). The class $\mathfrak{R}_{k,\ell}$ consists of unit-reset strategies for which, conditional on every recorded first history, the entire retained old quantum system has dimension at most k and the entire fresh reference accompanying the second input has dimension at most ℓ . The old system and the complete fresh input–reference system are in a tensor product at that cut. All other quantum registers are discarded before the second call; classical records may be retained without a cardinality bound. The initial first-call reference and temporary workspaces before this cut have no dimension restriction. The final measurement on the two retained receivers

is arbitrary. Mixed preparations, recorded normal instruments, public randomization, null histories, discarded terminal calls and finite-dimensional tester closure have the conventions of Definition 2.1. Denote the optimal reward by $P_{k,\ell}$.

In particular, this is not a bound on the largest workspace throughout an implementation. Enlarging the old output by keeping an instrument environment would violate the stated resource. The refinement in the next lemma records that environment's Kraus index *classically* instead, without retaining it coherently.

Lemma 6.2 (Dimension-preserving refinement). *For the optimization over $\mathfrak{R}_{k,\ell}$, the normal form (2.1)–(2.2) is exact with*

$$(6.1) \quad \text{rank } A_{yh} \leq k, \quad \text{rank } b_{yh} \leq \ell.$$

Every implemented competitor can be refined into such branches while preserving its score if the added classical indices are ignored in its decisions. Conversely every finite Gram collection satisfying these ranks and the common-barycenter constraints is realizable in $\mathfrak{R}_{k,\ell}$.

Proof. Purify the first input, retaining a reference whose relevant Schmidt support has dimension at most d_1 . A completely positive branch of the subsequent map from that support to $\mathbb{C}^k$ has a finite Kraus representation $X \mapsto \sum_q V_{yhq} X V_{yhq}^*$, with $V_{yhq} : \mathbb{C}^{r_0} \rightarrow \mathbb{C}^k$. Refine h to (h, q) , record q classically, and use the original fresh preparation and final effects. Summing over q recovers the old channel and its decision statistics. Thus $C_{yhq} = V_{yhq} C_0$ has rank at most k , and completeness still gives $\sum_{h,q} C_{yhq}^* C_{yhq} = \rho$ for every y . An original mixed first preparation is recovered by discarding its purification before this old-output cut; this discarding is part of that same completely positive branch, not an additional retained quantum register.

A mixed fresh state on $I_2 \otimes \mathbb{C}^\ell$ has a finite spectral mixture of normalized pure states in that same space. Sample and record its mixture index, use the corresponding vector, and ignore that index in the original decision. Its coefficient map $D : I_2 \rightarrow \mathbb{C}^\ell$ has Gram rank at most ℓ . Multiplying the old coefficient map by the square root of this mixing probability preserves the common Gram sum. This is a classical refinement, not a purification into a larger fresh reference. It is independent of the old quantum output conditional on the refined record. Countable outcomes are handled by the trace tails of Lemma 2.2. Public mixtures and tester limits preserve the resulting value bound.

Conversely choose $C_0^* C_0 = \rho$ on its Schmidt support. Factor each A_{yh} as $C_{yh}^* C_{yh}$ with $C_{yh} : I_1 \rightarrow \mathbb{C}^k$, and each b_{yh} as $D_{yh}^* D_{yh}$ with $D_{yh} : I_2 \rightarrow \mathbb{C}^\ell$. The maps $V_{yh} = C_{yh} C_0^+$ satisfy $\sum_h V_{yh}^* V_{yh} = I$ on that support. They give a recorded trace-preserving instrument. The normalized fresh vector $|D_{yh}\rangle$ is prepared independently, and polar isometries transfer the prescribed leaf POVMs to $\mathbb{C}^k \otimes \mathbb{C}^\ell$. Zero Gram branches are simply zero maps. The original unnormalized formulas define all null histories. $\square$

6.2. An attained hierarchy and its dual. Write $\mathcal{D}_d^{\leq q}$ for the compact set of density matrices of rank at most q . In the notation of (2.4), put

$$(6.2) \quad g_y^\ell(a) = \max_{b \in \mathcal{D}_2^{\leq \ell}} \Phi_y(a, b),$$

$$(6.3) \quad c_y^{k,\ell}(\rho) = \max \left\{ \sum_h \lambda_h g_y^\ell(a_h) : a_h \in \mathcal{D}_1^{\leq k}, \quad \lambda_h \geq 0, \quad \sum_h \lambda_h a_h = \rho \right\}.$$

Weights sum to one by taking traces. All density matrices have such a representation, since rank-one atoms are allowed.

Theorem 6.3 (Rank-resolved variational principle). *For every finite two-call experiment,*

$$(6.4) \quad \begin{aligned} P_{k,\ell} &= \max_{\rho \in \mathcal{D}_{d_1}} \sum_y c_y^{k,\ell}(\rho) \\ &= \min_{(H_y)} \lambda_{\max} \left(\sum_y H_y \right), \quad \text{tr}(H_y a) \geq g_y^\ell(a) \quad (a \in \mathcal{D}_{d_1}^{\leq k}), \end{aligned}$$

where the H_y are Hermitian. Both extrema are attained. At an optimal barycenter ρ , at most $(\text{rank } \rho)^2$ nonzero receiver outcomes after each first label suffice; a final quantum receiver of dimension $k\ell$ suffices. The values increase monotonically in either dimension, with

$$P_{1,d_2} = P_{\text{ca}}, \quad P_{d_1,d_2} = P_{\text{reset}}.$$

The contact, spectral and quantitative defect criteria of Theorem 2.5 and Corollary 2.6 hold with g_y^ℓ and the rank-restricted atoms.

Proof. The function Φ_y is continuous and homogeneous in its first positive argument. The rank-bounded sets are compact, so g_y^ℓ is continuous and its graph on $\mathcal{D}_{d_1}^{\leq k}$ has compact convex hull. The largest height over each ρ is an attained, upper semicontinuous concave envelope. Lemma 6.2 gives both operational directions. The atom-elimination proof of Theorem 2.4 uses only linear dependence of atoms on the support of ρ , and does not change any atom. It therefore preserves the rank restriction and gives the same branch bound. Compactness supplies all fresh states and final POVMs.

For clarity, dual attainment does not require the rank-bounded atom set to be convex. The homogeneous hypograph of its concave envelope is a closed convex cone, finite on every positive matrix and nonnegative there. At the common matrix I/d_1 with every height -1 , all these cones are strictly feasible. The conic argument of Theorem 2.5 applies without alteration: its dual is precisely majorization of the atom graph in (6.4). Bounded reward gives a finite value. Slater duality yields no gap and an attained Hermitian majorant. Subtracting the primal value from the dual upper gives the same three nonnegative defects, proving the contact and quantitative assertions. Inclusion of the rank-bounded atom sets proves monotonicity. The two endpoint identifications are Theorems 2.8 and 2.4. $\square$

The dual inequalities are universal on their rank-bounded domains; sampling is not a certificate. In particular, strict improvement between any two nested dimension pairs is equivalent to a feasible finite strategy for the larger pair exceeding an attained dual upper for the smaller pair. This criterion covers arbitrary fixed pairs, without asserting that every such pair exhibits a gap.

6.3. Rank-sensitive swap bounds. The rank in the next lemma is an upper bound on each of the two Gram ranks; the active overlap may have a smaller rank.

Lemma 6.4 (Swap bounds at finite rank). *Let $1 \leq r \leq d$, $d \geq 2$, and let $A, B \succeq 0$ have ranks at most r . With $L = \sqrt{A} \otimes \sqrt{B}$,*

$$(6.5) \quad \text{tr}[-LSL]_+ \leq \frac{r-1}{2r} \text{tr } A \text{tr } B,$$

$$(6.6) \quad \|L(I - dS)L\|_1 \leq \left(d - \frac{1}{r}\right) \text{tr } A \text{tr } B.$$

The bounds also hold if only the smaller of the two ranks is at most r , and for rectangular factors with the same Grams. Both bounds are sharp: take $A = aP/r$, $B = bP/r$, where P is any rank- r projection and $a, b \geq 0$. For nonzero factors, equality in the first bound with $r > 1$ requires these forms with a common P ; equality in the second requires them when $(d, r) \neq (2, 1)$.

Proof. The nonzero eigenvalues of $\sqrt{AB}\sqrt{A}$ number at most r . Consequently, for $f = \|\sqrt{A}\sqrt{B}\|_1$,

$$\mathrm{tr}(AB) \geq f^2/r, \quad f^2 \leq \mathrm{tr} A \mathrm{tr} B.$$

Insert the first inequality into Lemma 4.3 to get (6.5). The triangle step of Lemma 5.1 gives

$$\|L(I - dS)L\|_1 \leq \mathrm{tr} A \mathrm{tr} B - \mathrm{tr}(AB) + (d - 1)f^2 \leq \mathrm{tr} A \mathrm{tr} B + (d - 1 - r^{-1})f^2.$$

The coefficient is nonnegative, proving (6.6). These arguments only use the smaller rank, not a common support assumption. The spectrum identities hold for singular factors by the proved continuity argument, and polar isometries handle rectangular maps.

For equality, when the coefficient of f^2 is positive, normalized fidelity one gives $A/\mathrm{tr} A = B/\mathrm{tr} B$ without an invertibility assumption, as in Lemma 5.1. Equality in the rank- r Cauchy–Schwarz inequality then forces exactly r equal nonzero eigenvalues, proving the common-projection forms. For such forms the centered spectrum is $(1 - d)ab/r^2$ with multiplicity $r(r + 1)/2$ and $(1 + d)ab/r^2$ with multiplicity $r(r - 1)/2$; the negative swap has mass $ab(r - 1)/(2r)$. This proves sharpness. The excluded centered case has zero coefficient and different equality cases: for $d = 2, r = 1$, orthogonal pure factors also attain the bound. For the first inequality at $r = 1$, the negative mass is always zero. $\square$

6.4. The complete two-cut spectrum of the basis experiment. The following formula resolves every pair of retained-register thresholds on the same once-selected finite basis family. No additional design is introduced at a new threshold.

Theorem 6.5 (Exact dimension spectra). *Fix $d \geq 2$, $0 \leq t \leq 1$, and a weighted exact second-moment family satisfying (4.1). Under the once-drawn device rule, let $P_{k,\ell}^b$ use the priors of Theorem 4.2, and let $P_{k,\ell}^{\mathrm{eq}}$ use the equal hypothesis priors of Theorem 5.2. For $1 \leq k, \ell \leq d$, set $r = \min\{k, \ell\}$ and $L_0 = 2(d + 1) - t^2$. Then*

$$(6.7) \quad P_{k,\ell}^b = \frac{d + 1}{L_0} + \frac{t^2(r - 1)}{2rL_0},$$

$$(6.8) \quad P_{k,\ell}^{\mathrm{eq}} = \frac{1}{2} + \frac{t^2(dr - 1)}{2dr(d + 1)}.$$

These are optima over the complete classes $\mathfrak{R}_{k,\ell}$. For $t > 0$ they strictly increase at every increase of the smaller threshold, and attain the unrestricted reset values precisely at $k = \ell = d$. The two-cut resource dependence is therefore exactly through $\min\{k, \ell\}$ for these experiments.

Proof. Use the refined normal form of Lemma 6.2; there is no retained dilation environment outside the counted receivers. Every branch has $\mathrm{tr} b_{yh} = 1$ and smaller Gram rank at most r , while $\sum_{yh} \mathrm{tr} A_{yh} = d$. For the biased priors the equal-label payoff is $-t^2S/(dL_0)$ and all unequal-label payoffs are nonpositive. Equation (6.5) bounds the gain by $t^2(r - 1)/(2rL_0)$. For equal priors write $\alpha = t^2/[2d^2(d + 1)]$. The $d - 1$ opposite-sign payoff blocks cancel their denominator, so the branch gain is at most $\alpha\|(C_{yh} \otimes D_{yh})(I - dS)(C_{yh} \otimes D_{yh})^*\|_1$. Equation (6.6) and the Gram sum give $\alpha d(d - 1/r)$, which is (6.8) minus $1/2$. Classical mixtures and trace-norm closure preserve both upper bounds.

Here is a single finite construction attaining them. Index the input basis cyclically modulo d , and let $J_h : \mathbb{C}^r \rightarrow \mathbb{C}^d$ include the r consecutive coordinates starting at h , $0 \leq h < d$. Put $P_h = J_h J_h^*$; then $\sum_h P_h = rI$. Choose the first coefficient map $C_0 = I/\sqrt{d}$ and, after every first device label, the receiver instrument and fresh coefficient maps

$$(6.9) \quad V_h = J_h^*/\sqrt{r}, \quad C_{yh} = J_h^*/\sqrt{dr}, \quad D_{yh} = J_h^*/\sqrt{r}.$$

Thus $\sum_h V_h^* V_h = I$, $A_{yh} = P_h/(dr)$, $b_{yh} = P_h/r$ and $\sum_h A_{yh} = I/d$. The fresh vector has norm one and is independent of the old receiver. Both retained quantum registers have dimension r ; h

is classical. The initial reference has dimension d , as explicitly allowed before the recording cut in Definition 6.1.

On these receivers the filtered swap is $S_r/(dr^2)$ and the centered one is $(I_{r^2} - dS_r)/(dr^2)$. For the biased prior, declare C only on equal labels and the antisymmetric receiver projection. For equal priors, declare C on the antisymmetric projection on equal labels and on the symmetric projection on unequal labels. Complementary effects complete a POVM. Their spectral signs attain every branch bound above. At $r = 1$ the antisymmetric projection is zero; the second readout reduces to the unequal-label rule. At $r = d$ the instrument's redundant classical randomization may be omitted. All constructions remain normalized at $t = 0$ and on projective null histories at $t = 1$. $\square$

Corollary 6.6 (Adjacent gains and dimension certificates). *For $1 \leq r < d$, abbreviate the values in Theorem 6.5 by P_r^b and P_r^{eq} . Their adjacent gains are*

$$P_{r+1}^b - P_r^b = \frac{t^2}{2L_0r(r+1)}, \quad P_{r+1}^{\text{eq}} - P_r^{\text{eq}} = \frac{t^2}{2d(d+1)r(r+1)}.$$

For the equal-prior perturbations of Corollary 5.3, with its score error κ , every perturbed class value moves by at most κ . In particular an achieved ideal-reset score strictly above $P_r^{\text{eq}} + \kappa$ requires both counted receiver dimensions to exceed r . The adjacent gap survives if 2κ is smaller than its displayed value. A certified additional preparation defect ε changes the implemented score by at most $\varepsilon/2$ and may be added to these error budgets.

Proof. Subtract the consecutive exact formulas. The hybrid and probability law bounds are uniform in a fixed protocol, independently of its receiver dimensions, so they hold for every rank-restricted supremum. If either dimension were at most r , its nominal value would be at most P_r^{eq} by Theorem 6.5. This proves the asserted certificate and the perturbation statements. $\square$

These dimension certificates assume the specified reset architecture and calibration bounds. They are not device-independent conclusions from an uncalibrated score. The entire two-cut hierarchy still lies inside the reset class; the unrestricted coherent-acquisition optimum of Theorem 5.2 is a different resource.

6.5. Relation to constrained-memory discrimination. Memory-constrained tester optimization and its relation to constrained separability are established in [2], including the classical adaptive boundary. Theorem 6.3 is a rank-envelope characterization for the specified classical-output, fresh-preparation cut, and Theorem 6.5 solves its two-coordinate values on one fixed finite family. It does not replace the general channel-memory formulations of that work. The autonomous multi-time hierarchy of [3] carries coherent information between successive acquisitions and compiles arbitrary finite-time testers with a counter. Here increasing the two receiver dimensions never removes the reset cut. Consequently saturation means the full reset optimum, not the unrestricted strategy norm. These different saturation statements are compatible and cannot be interchanged.

7. INITIAL SPECTRA AND RIGIDITY OF RANK-OPTIMAL RECEIVERS

The dimension spectrum determines an optimal value. We now determine which initial spectra can attain it, and how a small loss in value restricts the acquisition marginals. Throughout this section the basis index is drawn once and retained for both calls, the hypothesis priors are equal, and $0 < t \leq 1$. The family satisfies (4.1). Put

$$r = \min\{k, \ell\}, \quad p_r = \frac{1}{2} + \frac{t^2(dr - 1)}{2dr(d + 1)}, \quad \alpha = \frac{t^2}{2d^2(d + 1)}.$$

All ranks and norms refer to the fixed input copies. Trace norms are unhalved. An initial Gram matrix ρ means a pure first acquisition $|C_0\rangle$ with $C_0^*C_0 = \rho$, in the convention of Lemma 2.3; its input marginal is $\rho^\top$. The receiver cut is exactly Definition 6.1.

7.1. The complete set of optimal initial spectra.

Theorem 7.1 (Optimal initial Gram matrices). *A fixed initial Gram matrix $\rho \in \mathcal{D}_d$ admits a strategy in $\mathfrak{R}_{k,\ell}$ attaining p_r if and only if*

$$(7.1) \quad \rho \preceq I/r.$$

Every such ρ has a decomposition

$$(7.2) \quad \rho = \sum_{h=1}^s w_h P_h / r, \quad s \leq d, \quad w_h > 0, \quad \sum_h w_h = 1,$$

where the P_h are rank- r projections commuting with ρ . This decomposition gives an attaining strategy with the same receiver instrument after every first device label and with both retained quantum registers of dimension r .

Except when $(d, r) = (2, 1)$, every attaining strategy in the dimension-preserving refined normal form has, on each nonzero branch,

$$(7.3) \quad A_{yh} = w_{yh} P_{yh} / r, \quad b_{yh} = P_{yh} / r, \quad \text{rank } P_{yh} = r.$$

Together with the common Gram sums and maximizing leaf decisions, these conditions are also sufficient. They specify Gram data, not unique dilations or unique final measurements.

Proof. For a normalized branch write $a = A / \text{tr } A$ and $b = b_{yh}$. The equal-prior upper in Theorem 6.5, before applying its final inequality, is

$$(7.4) \quad v \leq \frac{1}{2} + \alpha \sum_{y,h} w_{yh} \|(\sqrt{a_{yh}} \otimes \sqrt{b_{yh}})(I - dS)(\sqrt{a_{yh}} \otimes \sqrt{b_{yh}})\|_1.$$

The smaller Gram rank is at most r , and $\sum_{y,h} w_{yh} = d$. All branch deficits from the bound $d - 1/r$ are nonnegative. Attainment therefore forces equality on every branch of positive weight. When $(d, r) \neq (2, 1)$, Lemma 6.4 gives (7.3). For each fixed y , summing yields $\rho = \sum_h w_{yh} P_{yh} / r \preceq I/r$, since $\sum_h w_{yh} = 1$. In the exceptional case $r = 1$, the claimed spectral condition holds for every density matrix anyway. The fixed- ρ compact-envelope argument of Theorem 6.3 gives attainment of its constrained value, so the same conclusion applies when an optimum was initially defined through tester closure.

We give a finite construction for the converse. Diagonalize ρ with eigenvalues λ_i and set $x_i = r\lambda_i$ and $s_i = \sum_{j \leq i} x_j$, $s_0 = 0$. Then $0 \leq x_i \leq 1$ and $s_d = r$. For almost every $u \in [0, 1)$, mark the r points $u, u+1, \dots, u+r-1$ in $[0, r)$. Select coordinate i if its half-open interval $[s_{i-1}, s_i)$ contains a marked point. Each interval has length at most one, so exactly r distinct coordinates are selected. The probability of selecting coordinate i is x_i . The selected subset is constant between successive fractional parts of the s_i . There are at most d such intervals of positive length, because s_0 and s_d have the same fractional part. Their lengths are the weights w_h , and their selected coordinate projections are P_h . Taking diagonal entries proves (7.2). Endpoints have zero measure and may be assigned either adjacent choice. Rational eigenvalues give rational weights by this construction; no rational eigenbasis is presumed.

Choose isometries $J_h : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $J_h J_h^* = P_h$ and put

$$C_{yh} = \sqrt{w_h/r} J_h^*, \quad D_{yh} = J_h^* / \sqrt{r}, \quad V_{yh} = C_{yh} C_0^+.$$

On the initial Schmidt support, the common Gram sum gives $\sum_h V_{yh}^* V_{yh} = I$ and $V_{yh} C_0 = C_{yh}$. Thus V_{yh} is a recorded instrument, and $|D_{yh}\rangle$ is an independently prepared normalized fresh

vector. The same maps may be used for every y . The receiver centered swap is $w_h(I_{r^2} - dS_r)/r^2$. Declare the scalar hypothesis on the antisymmetric receiver projection for equal labels and on the symmetric projection for unequal labels. Complementary effects complete the measurement. These decisions attain each positive-part bound, proving sufficiency, including $r = 1$. Applying the same construction without a prescribed commuting eigenbasis proves sufficiency of (7.3) with the common Gram sums. Countable normal forms are covered by the positive trace tails of Lemma 2.2. $\square$

In particular the initial state need not be maximally mixed when $r < d$. Its largest Schmidt weight must be at most $1/r$. When $r = d$, the condition reduces to $\rho = I/d$. The finite interval construction concerns a known initial spectrum, not construction of the second-moment family itself.

7.2. Counting the first acquisition as well. Let $\mathfrak{R}_{q;k,\ell}$ impose the additional condition that the entire reference accompanying the first input has dimension at most q , conditional on the public preparation record. Mixed states on $I_1 \otimes \mathbb{C}^q$ are allowed. The old receiver after its recorded instrument has dimension at most k and the independent second reference at most ℓ , as before. Workspaces discarded before a specified cut do not enlarge a counted register. Classical records and the final joint receiver measurement have the same conventions as Definition 6.1.

Corollary 7.2 (Three specified dimension cuts). *For $1 \leq q, k, \ell \leq d$, put $s = \min\{q, k, \ell\}$. On the same finite family the complete classes $\mathfrak{R}_{q;k,\ell}$ have exact optimal values*

$$(7.5) \quad \begin{aligned} P_{q;k,\ell}^{\text{eq}} &= \frac{1}{2} + \frac{t^2(ds - 1)}{2ds(d + 1)}, \\ P_{q;k,\ell}^{\text{b}} &= \frac{d + 1}{2(d + 1) - t^2} + \frac{t^2(s - 1)}{2s[2(d + 1) - t^2]}. \end{aligned}$$

The second line uses exactly the prior of Theorem 4.2. Both optima are attained by two independently prepared maximally entangled vectors on any fixed rank- s input subspace, with no intervening receiver instrument and no feedback. No first-call reference of dimension d is required unless $s = d$.

Proof. Resolve a mixed first preparation into a finite pure-state mixture in $I_1 \otimes \mathbb{C}^q$ and record its mixture index classically. This recovers its old score if the index is ignored, without adding a quantum purification. For each component rank $C_0 \leq q$. The Kraus refinement of Lemma 6.2 then gives rank $A_{yh} \leq \min\{q, k\}$, while rank $b_{yh} \leq \ell$. The two rank-sensitive swap bounds apply with s . The trace sums, constant-decision baselines and positive-part optimizations in Theorem 6.5 give both upper bounds. Averaging over initial mixtures and public seeds, and then taking norm limits, preserves these bounds.

For attainment fix an isometry $J : \mathbb{C}^s \rightarrow \mathbb{C}^d$ and use $C_0 = D = J^*/\sqrt{s}$. Keep the first receiver unchanged and prepare the second vector freshly. For every first label the old Gram and the fresh Gram are both JJ^*/s . The two final spectral readouts used in Theorem 6.5 attain the bounds. All three counted registers have dimension s . This proves the assertion also at $s = 1$, where the antisymmetric receiver projection is zero. $\square$

The three bounds apply at specified cuts. They are not a cap on the total workspace of an arbitrary implementation or a removal of the reset architecture. In particular, saturation remains the reset value, not the unrestricted coherent-acquisition value.

7.3. A stable common-projection theorem. For density matrices a, b with smaller rank at most r , set

$$\Delta_{d,r}(a, b) = d - r^{-1} - \|(\sqrt{a} \otimes \sqrt{b})(I - dS)(\sqrt{a} \otimes \sqrt{b})\|_1.$$

When $(d, r) \neq (2, 1)$ define

$$(7.6) \quad c_{d,r} = d - 1 - r^{-1} > 0, \quad K_{d,r} = r + \frac{(3 + \sqrt{2})^2}{c_{d,r}}.$$

Lemma 7.3 (Quantitative common support). *For $d \geq 2$, $1 \leq r \leq d$ and $(d, r) \neq (2, 1)$, there exists a rank- r projection P such that*

$$(7.7) \quad \max\{\|a - P/r\|_1^2, \|b - P/r\|_1^2\} \leq K_{d,r} \Delta_{d,r}(a, b).$$

Only the smaller Gram rank need be bounded by r ; neither invertibility nor equality of the two supports is assumed.

Proof. Put $f = \|\sqrt{a}\sqrt{b}\|_1$, $\eta = 1 - f^2$ and $e = \text{tr}(ab) - f^2/r$. The overlap rank bound gives $e \geq 0$, and $f \leq 1$ gives $\eta \geq 0$. The proof of Lemma 6.4 gives

$$(7.8) \quad \Delta_{d,r}(a, b) \geq c_{d,r}\eta + e.$$

Extend a right polar factor to a unitary U so that $Z = \sqrt{a}\sqrt{b}U = \sqrt{\sqrt{ab}\sqrt{a}} \succeq 0$. The Hilbert–Schmidt identity and the Schatten product inequality give

$$\|\sqrt{b}U - \sqrt{a}\|_2^2 = 2(1 - f), \quad \|Z - a\|_1 \leq \sqrt{2(1 - f)} \leq \sqrt{2\eta}.$$

Choose any rank- r projection P containing the range of Z . This is possible since $\text{rank } Z \leq r$. Since $\text{tr } Z = f$, one has

$$\|Z - fP/r\|_2^2 = e, \quad \|Z - fP/r\|_1 \leq \sqrt{re}.$$

These formulas remain valid at $f = 0$, without normalizing Z . As $1 - f \leq \sqrt{\eta}$, the triangle inequality yields

$$\|a - P/r\|_1 \leq (1 + \sqrt{2})\sqrt{\eta} + \sqrt{re}.$$

The normalized coefficient vectors for $\sqrt{a}$ and $\sqrt{b}U$ have inner product f and receiver marginals a, b . Their pure-state trace distance is $2\sqrt{1 - f^2}$, by its rank-two spectral calculation. Contractivity under partial trace therefore gives $\|a - b\|_1 \leq 2\sqrt{\eta}$, the trace-distance/root-fidelity estimate of Fuchs–van de Graaf [9, Theorem 1]. Consequently

$$\|b - P/r\|_1 \leq (3 + \sqrt{2})\sqrt{\eta} + \sqrt{re}.$$

Weighted Cauchy–Schwarz bounds the square of either right side by $[r + (3 + \sqrt{2})^2/c_{d,r}](c_{d,r}\eta + e)$. Now use (7.8). Every argument used matrices on finite supports and a unitary extension of a partial isometry; no inverse of a singular matrix or limit of an equality statement occurs. $\square$

The constants in this lemma are explicit but not asserted to be optimal. The projection may vary between branches; the theorem does not force all receivers to occupy one common subspace throughout the protocol.

Corollary 7.4 (Near-optimal projection decompositions). *Assume $(d, r) \neq (2, 1)$ and a strategy in $\mathfrak{R}_{k,\ell}$ has reward $v \geq p_r - \varepsilon$. Choose a dimension-preserving refined normal form; on its nonzero branches put*

$$w_{yh} = \text{tr } A_{yh}, \quad a_{yh} = A_{yh}/w_{yh}, \quad \mu_{yh} = w_{yh}/d, \quad B_\varepsilon = \frac{2K_{d,r}d(d+1)}{t^2}\varepsilon.$$

There are rank- r projections P_{yh} such that

$$(7.9) \quad \sum_{y,h} \mu_{yh} (\|a_{yh} - P_{yh}/r\|_1^2 + \|b_{yh} - P_{yh}/r\|_1^2) \leq 2B_\varepsilon,$$

$$(7.10) \quad \frac{1}{d} \sum_y \left\| \rho - \sum_h w_{yh} P_{yh}/r \right\|_1^2 \leq B_\varepsilon.$$

Finite and countable recorded instruments are allowed. The weights μ_{yh} sum to one and are Gram weights, not probabilities under a chosen hypothesis. For public mixtures the same statements hold with the public seed adjoined to the refined record, using the common purified initial Gram matrix.

Proof. Subtract (7.4) from the attained class value. Since $\sum_{y,h} w_{yh} = d$, this gives

$$\sum_{y,h} \mu_{yh} \Delta_{d,r}(a_{yh}, b_{yh}) \leq \frac{\varepsilon}{\alpha d} = \frac{2d(d+1)}{t^2} \varepsilon.$$

Apply Lemma 7.3 to each branch and sum. For the second estimate, each fixed y has $\sum_h w_{yh} = 1$ and $\sum_h w_{yh} a_{yh} = \rho$. The triangle inequality followed by Jensen's inequality bounds its squared residual by $\sum_h w_{yh} \|a_{yh} - P_{yh}/r\|_1^2$. Average over y . The proof uses nonnegative sums with finite total weight, so countable traces pass to the limit. The seed construction in Lemma 2.3, followed by the dimension-preserving refinement, preserves the common Gram sum and the achieved reward. $\square$

The total μ -weight on branches for which either deviation is at least $u > 0$ is at most $2B_\varepsilon/u^2$. This is a quantitative statement at the relative loss scale ε/t^2 , not a claim of vanishing error at fixed nonzero calibration defect.

7.4. A spectral loss certificate. Let

$$\mathcal{C}_{d,r} = \{\sigma \in \mathcal{D}_d : \sigma \preceq I/r\}, \quad e_r(\rho) = \text{tr}(\rho - I/r)_+.$$

Proposition 7.5 (Distance to the optimal initial spectra). *For every density matrix ρ ,*

$$(7.11) \quad \inf_{\sigma \in \mathcal{C}_{d,r}} \|\rho - \sigma\|_1 = 2e_r(\rho).$$

For $(d,r) \neq (2,1)$, any strategy in $\mathfrak{R}_{k,\ell}$ with initial Gram ρ and achieved equal-prior reward v satisfies

$$(7.12) \quad p_r - v \geq \frac{2t^2}{K_{d,r}d(d+1)} e_r(\rho)^2.$$

Proof. Let Q be the spectral projection of ρ above $1/r$. For $\sigma \preceq I/r$, one has $\text{tr} Q(\rho - \sigma) \geq e_r(\rho)$. As $\text{tr}(\rho - \sigma) = 0$, positive-part optimization gives $\|\rho - \sigma\|_1 \geq 2e_r(\rho)$. Conversely, in an eigenbasis of ρ , cap every eigenvalue exceeding $1/r$ and redistribute the removed mass among eigenvalues below $1/r$. There is enough capacity since $d/r \geq 1$. This produces a density matrix in $\mathcal{C}_{d,r}$ at trace distance exactly $2e_r(\rho)$.

Every matrix $\sum_h w_{yh} P_{yh}/r$ in (7.10) belongs to $\mathcal{C}_{d,r}$. Set $\varepsilon = p_r - v \geq 0$ there and use (7.11); rearranging $4e_r(\rho)^2 \leq 2K_{d,r}d(d+1)(p_r - v)/t^2$ proves the bound. $\square$

The estimate concerns the actual first-input marginal, up to its fixed transpose, and a specified ideal reset architecture. Under the calibrated perturbations of Corollary 5.3, the same strategy's nominal reward differs from its observed reward by at most the stated channel, prior and latent-law error; a preparation defect adds its stated half-trace contribution. That total error must be added to $p_r - v$ before applying the certificate. There is no uncalibrated inference of the reset cut from a score.

7.5. The exceptional qubit face.

Proposition 7.6 (Pinching at the exceptional endpoint). *Let $d = 2$, $r = 1$, and suppose a is pure. Put $\mathcal{P}_a(b) = aba + (I - a)b(I - a)$ and $\Delta = \Delta_{2,1}(a, b)$. Then*

$$(7.13) \quad \|b - \mathcal{P}_a(b)\|_1^2 = 2\Delta - \Delta^2 \leq 2\Delta.$$

In particular $\Delta = 0$ if and only if $[a, b] = 0$. If instead b is the pure factor, interchange a, b .

Proof. By simultaneous unitary conjugation take $a = |0\rangle\langle 0|$ and $b = \begin{pmatrix} x & z \\ \bar{z} & 1-x \end{pmatrix}$. The nonzero block of the centered filtered swap is $\sqrt{b}(I - 2a)\sqrt{b}$. Its trace is $1 - 2x$ and its determinant is $-\det b$. Its trace norm, also when b is singular, is therefore

$$\sqrt{(1 - 2x)^2 + 4\det b} = \sqrt{1 - 4|z|^2} = 1 - \Delta.$$

The pinching difference has eigenvalues $\pm|z|$. Squaring gives (7.13). Vanishing is exactly $z = 0$, or $[a, b] = 0$. $\square$

Thus the exceptional equality set includes a pure factor with any commuting mixed qubit state, not just identical or orthogonal pure factors. A common rank-one projection cannot replace pinching here. The initial-spectrum theorem remains valid because $\mathcal{C}_{2,1} = \mathcal{D}_2$. This exception is separate from the zero-signal case $t = 0$, in which every strategy has reward $1/2$.

8. EXACT VALUES AT A PRESCRIBED INITIAL SPECTRUM

The equality set of a resource bound does not determine the value away from that set. We now solve this remaining optimization for the biased basis experiment of Theorem 4.2. Besides giving the exact spectral loss, the solution distinguishes protocols that merely retain a receiver from protocols that communicate a receiver outcome to the next preparation.

Fix $d \geq 2$, a weighted exact second-moment family satisfying (4.1), and $0 \leq t \leq 1$. The alternative basis is selected once and used at both calls. In this section only, the priors are those of (4.3); write

$$L_0 = 2(d + 1) - t^2, \quad p_E = (d + 1)/L_0.$$

Fix the *pure* first acquisition through its Gram matrix $\rho = C_0^* C_0 \in \mathcal{D}_d$. Its input marginal is $\rho^\top$. The initial acquisition is not replaced by a classically resolved mixture. Subsequent instruments, mixtures of fresh states, public seeds and tester limits retain the conventions of Definition 6.1.

For a nonnegative vector u , put $v_i = \sqrt{u_i}$ and define

$$(8.1) \quad \chi(u) = \lambda_{\max}(vv^* - \text{diag } u).$$

Appending zeros does not change this nonnegative number. In the secular equation omit zero coordinates; the matrix definition remains valid on every support face. If at least two entries of u are positive, it is the unique positive solution of

$$(8.2) \quad \sum_i \frac{u_i}{u_i + \chi} = 1.$$

If at most one entry is positive, set $\chi(u) = 0$.

Let $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ be the eigenvalues of ρ . For $1 \leq r \leq d$, choose the smallest $m \in \{0, \dots, r - 1\}$ for which

$$\lambda_{m+1} \leq \tau, \quad \tau = \frac{\sum_{i>m} \lambda_i}{r - m},$$

and define the rank- r majorant

$$(8.3) \quad q^{(r)}(\lambda) = (\lambda_1, \dots, \lambda_m, \underbrace{\tau, \dots, \tau}_{r-m}, 0, \dots, 0).$$

The choice always exists at $m = r - 1$. The preceding failed test, if $m > 0$, implies $\lambda_m > \tau$, so the displayed vector is decreasing. Zero tails are allowed; “rank- r ” here means rank at most r . Write $\mathcal{H}_r(\rho) = \chi(q^{(r)}(\lambda(\rho)))$. The “smallest m ” convention fixes the active set at a water-level tie. Entries equal to τ belong to the flattened tail. Including such a tied entry in the initial segment would give the same vector, so $q^{(r)}$ is unique even there. If r reaches the support size, $q^{(r)} = \lambda$, padded by zeros. These are the conventions for χ and $q^{(r)}$ throughout the article; eigenbasis choices are not part of their definitions. Lemma 8.5 proves the least-majorant assertion and hence uniqueness intrinsically.

Lemma 8.1 (Continuity at water-level ties). *On decreasing probability vectors the map $q^{(r)}$ is continuous, piecewise affine, and nonexpansive in ℓ^1 :*

$$\|q^{(r)}(\lambda) - q^{(r)}(\mu)\|_1 \leq \|\lambda - \mu\|_1.$$

Proof. For each fixed active index m , the displayed rule is an affine map: its first m coordinates are unchanged, and each remaining input coordinate contributes $1/(r - m)$ to each of the next $r - m$ outputs. Its matrix has nonnegative entries and every column sums to one, so its induced ℓ^1 norm is one. The finitely many closed active regions are polyhedral. On their overlaps the water-level equalities make the formulas identical, including zero tails. Hence the map is continuous. A line segment in the decreasing simplex has a finite partition into these regions; summing the norm bounds along its collinear subsegments proves the assertion. $\square$

Theorem 8.2 (Prescribed-spectrum value). *For the once-selected finite family and the fixed pure first acquisition above, the optimum over the complete class $\mathfrak{R}_{k,\ell}$, $1 \leq k, \ell \leq d$, is*

$$(8.4) \quad P_{k,\ell}^b(\rho) = p_E + \frac{t^2}{2L_0} \mathcal{H}_r(\rho), \quad r = \min\{k, \ell\}.$$

At $r = 1$ the spectral gain is zero. At $t = 0$ the value is $1/2$. If $r \geq \text{rank } \rho$, the majorant is the initial spectrum itself. Trace norms used in the derivation are unhalved; the factor $1/2$ in (8.4) converts the filtered-swap leaf normalization into this Bayes gain. The optimum is attained by an instrument with at most d recorded outcomes, the same instrument after every first device label, and fresh references of dimension at most r . Both retained quantum registers may have dimension r . The second preparation may depend on the recorded receiver outcome, but need not depend on the first device label. This construction acts on the prescribed initial acquisition; it does not optimize over its Gram matrix.

For $r = d$ no instrument is needed and the answer is given directly by the secular equation for $\lambda(\rho)$. The theorem gives an exact value for the specified priors. The equal-prior spectral rigidity of Section 7 remains a separate result.

8.1. The scalar function and a rank-constrained leaf.

Lemma 8.3 (Secular concavity). *The function χ is continuous, symmetric, homogeneous of degree one, coordinatewise nondecreasing and concave on each nonnegative orthant. It is strictly Schur-concave on each fixed positive trace simplex: if $u \prec w$ and u, w are not permutations, then $\chi(u) > \chi(w)$. Here vectors are padded to a common length.*

Proof. On a support containing at least two coordinates, the matrix in (8.1) has strictly positive off-diagonal entries. A maximizing Rayleigh vector can be chosen nonnegative, and its coordinates on this support must be positive. Its eigenvalue equation is $(u_i + \chi)x_i = \sqrt{u_i} \langle v, x \rangle$. The Rayleigh value is positive, and summing gives (8.2). Its left side strictly decreases from the support size to zero, proving uniqueness. The support-one case follows directly from the matrix formula. That formula also gives continuity, symmetry and homogeneity, including at the boundary.

For $a > 0$, the root characterization says

$$\chi(u) \geq a \iff \sum_i \frac{u_i}{u_i + a} \geq 1.$$

Each summand is increasing and concave in u_i , so every such superlevel set is convex and coordinatewise increasing. If $c = \chi(u) > 0$ and $e = \chi(w) > 0$, convexity of the unit superlevel set applied with weights $c/(c+e), e/(c+e)$ gives $\chi(u+w) \geq c+e$. If one value vanishes the same inequality follows from coordinate monotonicity. Homogeneity now proves concavity.

Symmetry and concavity imply Schur concavity by convex combinations of permutations. For strictness, if $\chi(w) > 0$, strict concavity of $x/(x + \chi(w))$ shows that $u \prec w$, without permutation equality, implies $\sum_i u_i/(u_i + \chi(w)) > 1$. The root for u is therefore larger. If $\chi(w) = 0$, w has only one positive entry and any nonpermutation vector majorized by it has at least two, again giving strictness. $\square$

For $A, B \succeq 0$ let

$$\mathcal{N}(A, B) = \text{tr}[-(\sqrt{A} \otimes \sqrt{B})S(\sqrt{A} \otimes \sqrt{B})]_+.$$

Lemma 8.4 (Optimal fresh Gram at a fixed old Gram). *If $a_1 \geq \dots \geq a_d \geq 0$ are the eigenvalues of A , then*

$$(8.5) \quad \max_{B \in \mathcal{D}_d, \text{rank } B \leq \ell} \mathcal{N}(A, B) = \frac{1}{2} \chi(a_1, \dots, a_\ell).$$

A maximizing B commutes with A and is supported on its leading ℓ eigenspaces. If $c = \chi(a_1, \dots, a_\ell) > 0$, an explicit choice on that support is

$$(8.6) \quad b_i = \frac{a_i/(a_i + c)^2}{\sum_{j \leq \ell} a_j/(a_j + c)^2}.$$

No invertibility of A is required. A commuting optimizer exists; if a leading eigenspace is degenerate the chosen eigenbasis, and hence the displayed optimizer, need not be unique. The factor $1/2$ in (8.5) concerns negative mass, with no halving of the trace norm in Lemma 4.3.

Proof. Lemma 4.3 expresses $2\mathcal{N}(A, B)$ as $(\text{tr } \sqrt{Z})^2 - \text{tr } Z$, where $Z = \sqrt{A}B\sqrt{A}$. First work on the support of A . Compressing B to that support leaves Z unchanged and does not increase its rank. If the compressed trace is positive, normalizing it can only increase the nonnegative, homogeneous objective. If the trace vanishes the objective is zero. We may therefore take A invertible on its support and $\text{tr}(A^{-1}Z) = 1$.

For fixed eigenvalues $z_1 \geq \dots \geq z_s > 0$ of Z , $s \leq \ell$, rearrangement gives

$$\sum_{i=1}^s z_i/a_i \leq \text{tr}(A^{-1}Z) = 1.$$

For completeness, in eigenbases the trace is a sum of products weighted by squared entries of a unitary, a doubly stochastic matrix. Its minimum over that larger convex set is achieved at a permutation; exchanging two inverted pairs puts the largest z_i on the largest a_i . This proves the inequality. Put Z' diagonal with these z_i on the leading eigenspaces of A and $B' = A^{-1/2}Z'A^{-1/2}$. Then $0 < \text{tr } B' \leq 1$; normalizing B' proves that a commuting maximizer with the asserted support suffices.

For such a B , write $x_i = \sqrt{b_i}$, so $\|x\|_2 = 1$. The objective is

$$2\mathcal{N}(A, B) = \left(\sum_{i \leq \ell} \sqrt{a_i} x_i \right)^2 - \sum_{i \leq \ell} a_i x_i^2.$$

The Rayleigh maximum is (8.1), with a nonnegative maximizer. Its eigenvector equation yields (8.6). With at most one positive a_i the maximum is zero, achieved by any rank-one Gram on the support, or any state if $A = 0$. This handles every singular case explicitly. $\square$

8.2. A spectral evaluation of the concave envelope. The next lemma records the majorization step separately. It applies to every continuous symmetric concave function, not only to the leaf objective just derived. The underlying majorization and ensemble methods are classical; their operational use in entanglement transformation and assistance is discussed in [10, 11, 12].

Lemma 8.5 (Rank-capped spectral envelopes). *Let f be a continuous symmetric concave function on the probability simplex of length r . For a density matrix of rank at most r , pad its eigenvalue vector to length r before applying f . Then*

$$(8.7) \quad \max_{\substack{\rho = \sum_h w_h a_h, \ w_h \geq 0 \\ a_h \in \mathcal{D}_d^{\leq r}}} \sum_h w_h f(\lambda(a_h)) = f(q_1^{(r)}(\lambda(\rho)), \dots, q_r^{(r)}(\lambda(\rho))).$$

The maximum admits at most d atoms commuting with ρ , all with the same spectrum $q^{(r)}$. In particular $\mathcal{H}_r$ is concave on $\mathcal{D}_d$; its homogeneous extension to positive matrices is concave and superadditive.

Proof. Write $q = q^{(r)}(\lambda)$. Its construction gives $\lambda \prec q$. Indeed its initial m partial sums agree with those of λ ; each subsequent entry of λ is at most τ , and its total sum is one. Moreover q is majorized by every decreasing probability vector z of rank at most r which majorizes λ . Up to m this follows from the majorization inequalities. Between m and r the decreasing entries of z give

$$\sum_{i \leq j} z_i \geq \frac{r-j}{r-m} \sum_{i \leq m} z_i + \frac{j-m}{r-m} \geq \sum_{i \leq m} \lambda_i + (j-m)\tau = \sum_{i \leq j} q_i.$$

After r both totals are one. Thus q is the least such majorant.

For any finite decomposition, the variational formula for the sum of the j largest eigenvalues gives

$$\lambda(\rho) \prec z := \sum_h w_h \lambda(a_h),$$

where the eigenvalue vectors are decreasing and padded to length d . The vector z has rank at most r . Concavity and Schur concavity give $\sum_h w_h f(\lambda(a_h)) \leq f(z) \leq f(q)$.

For the converse, $\lambda \prec q$ implies that λ lies in the convex hull of the permutations of q . One direct verification is as follows. For every real vector b , rearrangement followed by summation by parts in the partial-sum inequalities gives $b \cdot \lambda \leq b^\downarrow \cdot \lambda \leq b^\downarrow \cdot q = \max_\pi b \cdot \pi q$. A separating hyperplane is therefore impossible. This convex hull lies in a hyperplane of dimension at most $d-1$, so Carathéodory reduces the combination to at most d permutations. In an eigenbasis of ρ they give commuting density matrices a_h of spectrum q . Their f values coincide, proving attainment.

Concavity follows also by combining any two attaining decompositions. For a countable decomposition, the same upper follows from the Ky Fan inequalities and Jensen's inequality by convergence of its bounded trace weights. At zero trace define the homogeneous extension to be zero. Trace-weighted convex combinations prove its concavity and superadditivity. Apply this with $f = \chi$ for the final assertion. $\square$

For clarity, the homogeneous extension just proved is

$$\hat{\mathcal{H}}_r(A) = \begin{cases} (\operatorname{tr} A) \mathcal{H}_r(A / \operatorname{tr} A), & \operatorname{tr} A > 0, \\ 0, & A = 0. \end{cases}$$

For positive A, B of traces $a, b > 0$, concavity on densities gives $\widehat{\mathcal{H}}_r(A + B) \geq \widehat{\mathcal{H}}_r(A) + \widehat{\mathcal{H}}_r(B)$ by weights $a/(a + b)$ and $b/(a + b)$; zero traces are immediate. Subsequent formulas write $\mathcal{H}_r(A)$ for this extension on positive matrices.

8.3. Proof of the prescribed-spectrum theorem.

Proof of Theorem 8.2. Use the dimension-preserving normal form of Lemma 6.2. The fixed initial Gram gives $\sum_h A_{yh} = \rho$ for every y . Put $w_{yh} = \text{tr } A_{yh}$ and $a_{yh} = A_{yh}/w_{yh}$ on nonzero branches. These are Gram weights; they are not substituted for the hypothesis-dependent history probabilities. Mixed fresh states and receiver channels are refined classically, without increasing either retained quantum dimension.

The biased payoff blocks are $-\kappa S$ for equal labels and $-2\kappa\Pi_-(d - 1) \preceq 0$ for unequal labels, where $\kappa = t^2/(dL_0)$. Thus only equal labels can improve on the constant alternative decision. Lemma 8.4 gives

$$P - p_E \leq \frac{\kappa}{2} \sum_{y,h} \chi(\lambda_1(A_{yh}), \dots, \lambda_\ell(A_{yh})).$$

Each summand is at most $\mathcal{H}_r(A_{yh})$. If $k \leq \ell$, the rank of A_{yh} is at most $r = k$ and the two expressions agree. If $\ell < k$, the first $r = \ell$ coordinates of its rank- r majorant are coordinatewise at least its leading r eigenvalues; use the monotonicity of χ . Superadditivity then gives

$$P - p_E \leq \frac{\kappa}{2} \sum_y \mathcal{H}_r(\rho) = \frac{t^2}{2L_0} \mathcal{H}_r(\rho).$$

The same bound survives public mixing, countable normal instruments and tester closure, by the fixed-Gram normal form and trace tails. At $t = 0$ all devices coincide and both sides reduce to $1/2$.

For attainment take the commuting decomposition $\rho = \sum_h w_h a_h$ supplied by Lemma 8.5, all a_h with spectrum $q^{(r)}$. On each support choose rectangular $C_h, D_h : \mathbb{C}^d \rightarrow \mathbb{C}^r$ with

$$C_h^* C_h = w_h a_h, \quad D_h^* D_h = b_h,$$

where b_h is the permuted optimizing fresh Gram in (8.6). If $\chi(q^{(r)}) = 0$, use any pure fresh Gram and the constant decision. Otherwise b_h is a density matrix of rank at most r , supported with a_h . The maps $V_h = C_h C_0^+$ satisfy $V_h C_0 = C_h$ and $\sum_h V_h^* V_h = I$ on the initial Schmidt support: positivity of the Gram sum implies that every C_h vanishes on $\ker \rho$. They therefore form an instrument on the given initial receiver. Proposition 9.1 gives its domains, codomains, completion on an unused orthogonal complement and the at-most- d message bound explicitly, without retaining an extra environment.

Use this same instrument after every y and prepare $|D_h\rangle$ independently conditional on its recorded outcome. For equal labels, the positive spectral projection of the filtered negative swap attains $\kappa w_h \chi(q^{(r)})/2$. For unequal labels choose the zero scalar-hypothesis effect. Complements complete each binary readout. Summing the d first labels gives equality. These unnormalized formulas also define the endpoint $t = 1$ and null histories. The construction proves attainment without an extra retained environment. $\square$

8.4. When a receiver message is strictly useful. The following comparison fixes the initial acquisition and changes only the permission to send a receiver outcome to the fresh source. Let $P_{d,r}^{\text{nm}}(\rho)$ be the optimum with old receiver dimension d and fresh reference dimension r , when the fresh state may depend on the first device label and an independent public seed, but on no outcome obtained by measuring the old receiver. Arbitrary receiver processing and the final joint readout remain allowed. Any old processing without such communication can be postponed to the final readout by Lemma 3.1; the old bound d contains the entire initial Schmidt support. This is a subclass of $\mathfrak{R}_{d,r}$, not the complete-measure-and-reprepare class.

Corollary 8.6 (Exact value of a receiver message). *For the fixed experiment and initial Gram above,*

$$(8.8) \quad P_{d,r}^{\text{nm}}(\rho) = p_E + \frac{t^2}{2L_0} \chi(\lambda_1, \dots, \lambda_r).$$

If $t > 0$, then

$$(8.9) \quad P_{d,r}^{\text{b}}(\rho) > P_{d,r}^{\text{nm}}(\rho) \iff 2 \leq r < \text{rank } \rho.$$

The strict gain is realized with at most d receiver outcomes and without first-device-label dependence of the fresh state.

Proof. Without a receiver message each first-label branch retains Gram ρ . Lemma 8.4 gives the displayed upper, and the commuting optimizer with the positive-part final decisions attains it. Public mixtures cannot improve an already maximized linear score. If $r = 1$, both secular values vanish. If $r \geq \text{rank } \rho$, no tail is moved and $q^{(r)} = \lambda$. In the remaining case the leading r eigenvalues are all positive, and $q^{(r)}$ increases at least one of them without decreasing any. For a vector with at least two positive entries the root strictly increases when any positive coordinate is increased, as follows immediately from (8.2). This proves both directions. $\square$

For an explicit example take $d = 3$, $r = 2$ and $\rho = \text{diag}(3/4, 1/8, 1/8)$. Then

$$q^{(2)} = (3/4, 1/4, 0), \quad \chi(q^{(2)}) = \sqrt{3}/4, \quad \chi(\lambda_1, \lambda_2) = \sqrt{6}/8.$$

With $L_0 = 8 - t^2$, the exact advantage of the message is

$$(8.10) \quad \frac{t^2(2\sqrt{3} - \sqrt{6})}{16L_0} > 0 \quad (t > 0).$$

Use two Gram atoms $a_1 = \text{diag}(3/4, 1/4, 0)$ and $a_2 = \text{diag}(3/4, 0, 1/4)$ with weights $1/2$. With $C_0 = \sqrt{\rho}$, explicit old-receiver maps are

$$V_1 = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

They satisfy $\sum_h V_h^* V_h = I_3$ and $C_h = V_h C_0$, $C_h^* C_h = a_h/2$. The fresh Grams are respectively $\text{diag}(1/2, 1/2, 0)$ and $\text{diag}(1/2, 0, 1/2)$; normalized fresh vectors have the corresponding two equal Schmidt weights. The final positive-part readouts used in the proof attain the claimed scores. Thus the improvement is a realizable recorded instrument, not a change of initial state or postselection on a favorable history.

Corollary 8.7 (Spectral loss and exact saturation). *For $r > 1$, the difference from the optimized rank- r value of Theorem 6.5 is exactly*

$$(8.11) \quad P_{k,\ell}^{\text{b}} - P_{k,\ell}^{\text{b}}(\rho) = \frac{t^2}{2L_0} \left(\frac{r-1}{r} - \mathcal{H}_r(\rho) \right).$$

At $t > 0$ it vanishes exactly when $\rho \preceq I/r$. For $t > 0$ and a fixed ρ of rank s , its constrained optimum strictly increases as the smaller receiver dimension increases from 1 to s , and is constant thereafter. The smallest dimension attaining its full-reset value is therefore s , including the trivial case $s = 1$. At $t = 0$ every dimension gives the same value $1/2$.

Proof. The uniform vector of length r is majorized by every probability vector of that length. Lemma 8.3 gives $\chi(q) \leq (r-1)/r$, with equality only at that uniform vector. The construction of $q^{(r)}$ makes this equivalent to $\lambda_1 \leq 1/r$. Subtracting (8.4) from (6.7) proves the exact loss identity.

The least-majorant property gives $q^{(r+1)} \prec q^{(r)}$. If $r < s$, these have respectively $r+1$ and r positive entries and are not permutations, so strict Schur concavity gives a strict increase. At $r \geq s$ both vectors equal $\lambda(\rho)$. This proves the saturation statement. $\square$

The secular root and the majorant are functions of a supplied spectrum. Bisection of (8.2) evaluates the value without an optimization over physical protocols; a finite convex decomposition then realizes it. Real input spectra and real weights are permitted. These statements do not assume a rational eigenbasis or give an efficient construction of the unknown-device design. In particular, the general gain from a receiver message proved here is compatible with the no-feedback globally attaining preparations in Corollary 7.2: those preparations optimize the initial Gram, whereas (8.9) holds it fixed.

9. FINITE ATTAINMENT ON A PRESCRIBED RECEIVER

We make the spectral converse operational on the actual initial Schmidt support. This also records all normalization factors for a three-dimensional example. Neither construction replaces a fixed pure initial acquisition by an announced mixture.

Proposition 9.1 (Support-exact finite instrument). *Let $C_0 : \mathbb{C}^d \rightarrow R$ satisfy $C_0^* C_0 = \rho$, $\text{tr } \rho = 1$, and let $S_0 = \text{ran } C_0 \subseteq R$. Suppose $\rho = \sum_{h=1}^n w_h a_h$, where $w_h > 0$, $\sum_h w_h = 1$ and $\text{rank } a_h \leq r$. For each h let b_h be a density matrix of rank at most ℓ . Choose coefficient maps*

$$C_h : \mathbb{C}^d \rightarrow \mathbb{C}^r, \quad C_h^* C_h = w_h a_h, \quad D_h : \mathbb{C}^d \rightarrow \mathbb{C}^\ell, \quad D_h^* D_h = b_h.$$

The Moore–Penrose inverse $C_0^+ : R \rightarrow \mathbb{C}^d$, zero on $S_0^\perp$, gives $V_h = C_h C_0^+ : R \rightarrow \mathbb{C}^r$ with

$$(9.1) \quad V_h C_0 = C_h, \quad \sum_h V_h^* V_h = P_{S_0}.$$

These maps extend to an instrument on R with the same n classical outcomes and quantum outputs of dimension at most r . With the fresh vector $|D_h\rangle$ prepared independently after outcome h , the receiver block is the product in (2.2). For the spectral decomposition used in Theorem 8.2, $n \leq d$ suffices.

Proof. If $x \in \ker \rho$, positivity and the barycenter equation imply $\langle x, a_h x \rangle = 0$ for every h . Hence $C_h x = 0$ and $C_h P_{\text{supp } \rho} = C_h$. The Moore–Penrose identities then give $V_h C_0 = C_h$ and

$$\sum_h V_h^* V_h = (C_0^+)^* \rho C_0^+ = P_{S_0}.$$

This proves (9.1) even when ρ is singular. To complete the instrument without adding a recorded outcome, choose a unit vector $e \in \mathbb{C}^r$ and add to its first branch the completely positive map

$$X \mapsto \text{tr}(P_{S_0^\perp} X) |e\rangle\langle e|.$$

Together with $X \mapsto V_1 X V_1^*$ in that branch and $X \mapsto V_h X V_h^*$ in the others, the total map is trace preserving. This completion vanishes on all physically occupied initial receiver states. It is normal also on a larger separable receiver space, and retains no dilation environment. The fresh vector has squared norm $\text{tr } b_h = 1$. Measurement of each input gives the product block by expansion in the fixed input bases. The final POVMs therefore implement the chosen leaf optima.

For the spectral theorem, put $q = q^{(r)}(\lambda(\rho))$. The least-majorant proof in Lemma 8.5 gives $\lambda(\rho) \prec q$. A decreasing vector majorized by q lies in the convex hull of its permutations; this follows by successive two-coordinate averaging, or equivalently by the doubly stochastic characterization of majorization. Every permutation belongs to the affine hyperplane $\sum_i x_i = 1$, of dimension $d - 1$. Carathéodory therefore reduces its convex combination to at most d terms. Use those permutations as diagonal atoms in a fixed eigenbasis of ρ and apply the construction above. Eigenvalue ties can change the chosen eigenbasis or decomposition, not the value or the bound. The weights may be real; this argument asserts no bit or gate complexity bound. $\square$

The probability of (y, h) under a first measurement E^θ is $w_h \operatorname{tr}[a_h(E_y^\theta)^\top]$. Thus w_h is generally not the conditional receiver-outcome probability after observing y . No physical branch is postselected in the attaining construction.

9.1. One complete three-dimensional calculation. Let $d = 3$, $r = 2$, $t = 1$, and

$$\rho = \operatorname{diag}(3/4, 1/8, 1/8), \quad |C_0\rangle = \frac{\sqrt{3}}{2}|1, 1\rangle + \frac{|2, 2\rangle + |3, 3\rangle}{2\sqrt{2}}.$$

Its squared norm is one. For a fully specified finite device family, put $\omega = e^{2\pi i/3}$. Use the coordinate basis and the three bases

$$|u_j^{(a)}\rangle = \frac{1}{\sqrt{3}} \sum_{x=0}^2 \omega^{ax^2+jx} |x+1\rangle, \quad a = 0, 1, 2, \quad j = 0, 1, 2,$$

with all six orders of each basis. These 24 ordered bases have weights $1/24$. Their projections satisfy (4.1). To check the equal-label identity directly, consider the 12 vectors before ordering. For their nine chirp vectors the $(x, u; x', u')$ tensor entry in the sum of projectors squared is

$$\frac{1}{9} \sum_{a,j=0}^2 \omega^{a(x^2+u^2-x'^2-u'^2)+j(x+u-x'-u')}.$$

It equals one exactly when the two multisets $\{x, u\}$ and $\{x', u'\}$ agree, and zero otherwise: equal sums and square sums over $\mathbb{F}_3$ imply equal products. The three coordinate vectors add one to the fully diagonal entries. The full sum is therefore $I + S$. Dividing by 12 proves the equal-label moment; summing the other two labels gives $(3I - S)/24$ for distinct labels. This supplies the exact weighted family, rather than only its existence.

Under the scalar hypothesis $C_y = I/3$. Under the alternative one of the 24 ordered projective measurements is drawn *once* and used twice. The priors are $p_C = 3/7$, $p_E = 4/7$, so each individual alternative has unconditional prior $1/42$. All these weights are rational, whereas the vectors and values below are algebraic.

Proposition 9.2 (A prescribed-spectrum message witness). *For this experiment and initial acquisition the exact communicated and no-message values with old dimension three and fresh dimension two are*

$$(9.2) \quad P_{3,2}^b(\rho) = \frac{4}{7} + \frac{\sqrt{3}}{56}, \quad P_{3,2}^{\text{nm}}(\rho) = \frac{4}{7} + \frac{\sqrt{6}}{112}.$$

Two receiver outcomes attain the first value using an old output of dimension two. The second value retains the complete old receiver and is an upper over all no-message strategies, not just the exhibited one. The strict gain is $(2\sqrt{3} - \sqrt{6})/112$.

Proof. The water-filling rule has $m = 1$, $\tau = 1/4$, hence $q^{(2)} = (3/4, 1/4, 0)$. Its secular root is $\sqrt{3}/4$, whereas the leading two original eigenvalues have root $\sqrt{6}/8$. For two positive entries (a, b) the secular equation reduces to $x^2 = ab$. In particular the disjoint rational enclosures

$$\frac{433}{1000} < \frac{\sqrt{3}}{4} < \frac{434}{1000}, \quad \frac{306}{1000} < \frac{\sqrt{6}}{8} < \frac{307}{1000}$$

are certified by squaring their positive endpoints. They are intervals for algebraic numbers, not exact rational values. Theorem 8.2 and Corollary 8.6, whose no-message reduction is Lemma 3.1, give both upper bounds in (9.2).

For explicit attainment use

$$a_1 = \operatorname{diag}(3/4, 1/4, 0), \quad a_2 = \operatorname{diag}(3/4, 0, 1/4), \quad w_1 = w_2 = 1/2,$$

so $\rho = (a_1 + a_2)/2$. With $C_0 = \sqrt{\rho} : \mathbb{C}^3 \rightarrow \mathbb{C}^3$, set

$$\begin{aligned} V_1 &= \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & D_1 &= \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 1/\sqrt{2} & 0 \end{pmatrix}, \\ V_2 &= \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, & D_2 &= \begin{pmatrix} 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} \end{pmatrix}. \end{aligned}$$

Every displayed map has domain $\mathbb{C}^3$ and codomain $\mathbb{C}^2$. The old coefficients are $C_h = V_h C_0$, and directly

$$\sum_h V_h^* V_h = I_3, \quad C_h^* C_h = a_h/2, \quad \text{tr } D_h^* D_h = 1.$$

Prepare $|D_h\rangle$ independently given the receiver outcome. On $\mathbb{C}^2 \otimes \mathbb{C}^2$ let $|\psi^-\rangle = (|1, 2\rangle - |2, 1\rangle)/\sqrt{2}$. For $y = z$ declare C on $|\psi^-\rangle\langle\psi^-|$; for $y \neq z$ its effect is zero. Use the complement for the alternative. Each branch filtered swap has negative eigenvalue $-\sqrt{3}/16$, with eigenvector $|\psi^-\rangle$. The payoff coefficient is $\kappa = 1/21$, so the total improvement, with three first labels and two outcomes, is $3 \cdot 2 \cdot \sqrt{3}/(21 \cdot 16) = \sqrt{3}/56$. The same instrument is used after every first label.

Without a message keep C_0 and use D_1 after every first label. The negative eigenvector is the antisymmetric vector in the first two old coordinates and the two fresh coordinates; its eigenvalue is $-\sqrt{6}/16$. The same equal-label readout gives improvement $3 \cdot \sqrt{6}/(21 \cdot 16) = \sqrt{6}/112$. The squared-norm normalizations above, not an unnormalized Bell vector, determine these factors. All outcomes are included and no initial mixture or favorable-history selection has been introduced. $\square$

The same construction with $0 < t < 1$ uses full-rank devices; its gain is $t^2(2\sqrt{3} - \sqrt{6})/[16(8 - t^2)]$. At $t = 0$ it is zero and both values equal $1/2$. A positive gain at this fixed ρ does not contradict the globally optimized no-message attainments at other initial spectra. The retained-register interpretation presupposes the specified calibrated reset cut.

10. EQUAL-PRIOR VALUES WITH A TWO-DIMENSIONAL RETAINED RECEIVER

The prescribed-spectrum formula in Section 8 uses the biased prior of the first basis experiment. Equal priors change not only the baseline but also the optimal fresh acquisition. We now solve the equal-prior problem at every pure initial acquisition when the old receiver has dimension at most two. The input dimension is arbitrary, and allowing a larger fresh reference does not change the answer. The unrestricted qubit instance consequently admits a complete initial-spectrum formula, including an optimal protocol without a receiver message.

Use a weighted exact second-moment family as in (4.1). With probability $1/2$ the device is $C_y = I/d$; otherwise draw b with law w_b *once*, and use $E_y^b(t) = (1 - t)I/d + tP_y^b$ at both calls. Here $d \geq 2$ and $0 \leq t \leq 1$. The initial acquisition is the prescribed pure vector $|C_0\rangle$, with Gram matrix $\rho = C_0^* C_0 \in \mathcal{D}_d$; its input marginal is $\rho^\top$. It is not replaced by a classically resolved initial mixture. All instruments, discarded calls, public records, null histories and norm limits follow Definition 6.1. Thus the complete retained old system, not merely one of its subsystems, has dimension at most two.

For $0 \leq v \leq 1$ define

$$\begin{aligned} c_d(v) &= d(d - 2) + (2d - 1)v, \\ (10.1) \quad \psi_d(v) &= d - 1 + \frac{(2d - 1)v}{2(d - 1 + d\sqrt{c_d(v)/(d^2 - 1)})}. \end{aligned}$$

The denominator is positive, including at $d = 2, v = 0$. Set

$$q = \max\{\lambda_{\max}(\rho), 1/2\}, \quad v_\rho = 4q(1 - q).$$

The pair $(q, 1 - q)$ is the nonzero part, with a possible zero entry, of the least rank-two majorant (8.3).

Theorem 10.1 (Equal-prior prescribed-spectrum value). *For the once-selected experiment and the fixed pure first acquisition above, and every $2 \leq \ell \leq d$, the full class $\mathfrak{R}_{2,\ell}$ has exact optimum*

$$(10.2) \quad P_{2,\ell}^{\text{eq}}(\rho) = \frac{1}{2} + \frac{t^2}{2d(d+1)} \psi_d(v_\rho).$$

A recorded instrument with at most d outcomes attains this value on the prescribed initial receiver. The same instrument may be used after every first device label; both retained quantum registers may have dimension two. The fresh preparation depends only on its recorded receiver outcome, not on the first device label. Its Schmidt weights are specified in Lemma 10.3 below.

For $t > 0$ the value equals the optimized rank-two benchmark precisely when $\lambda_{\max}(\rho) \leq 1/2$, and decreases strictly as that largest eigenvalue increases above $1/2$. At $t = 0$ every value is $1/2$. If either retained register has dimension at most one, the exact value at every fixed ρ is instead

$$(10.3) \quad \frac{1}{2} + \frac{t^2(d-1)}{2d(d+1)}.$$

The first assertion covers all fresh dimensions at least two while fixing the old dimension at two. It makes no assertion that an old receiver of larger dimension has the same value. The general-rank biased formula and the equal-prior globally optimized dimension hierarchy retain their respective quantifiers. If b is redrawn independently at the two calls, the averaged alternative is $\mathcal{M}_C \otimes \mathcal{M}_C$, and all equal-prior values in this section are $1/2$ instead.

10.1. A concavity principle for filtering. For a Hermitian X and a positive matrix M on a finite-dimensional space, let $\mathcal{F}_X(M) = \|\sqrt{M}X\sqrt{M}\|_1$.

Lemma 10.2 (Concavity in the filter Gram). *For every $M \succeq 0$,*

$$(10.4) \quad \mathcal{F}_X(M) = \max_{-M \preceq Z \preceq M} \text{tr}(XZ).$$

In particular $\mathcal{F}_X$ is continuous, positively homogeneous and concave on the positive cone. Consequently, for fixed $A \succeq 0$,

$$B \mapsto \|(\sqrt{A} \otimes \sqrt{B})(I - dS)(\sqrt{A} \otimes \sqrt{B})\|_1$$

is concave. If A is diagonal, replacing B by its diagonal pinching cannot decrease this last quantity.

Proof. Every Hermitian contraction F gives $Z = \sqrt{M}F\sqrt{M}$ in the interval $[-M, M]$. Conversely any matrix in this interval vanishes on $\ker M$. Indeed $M \pm Z \succeq 0$ have zero quadratic form on that kernel, so both annihilate it. On $\text{supp } M$, $F = M^{-1/2}ZM^{-1/2}$ is a Hermitian contraction; extend it by zero. The variational formula for the trace norm and cyclicity of the trace prove (10.4), without invertibility of M . Convex combinations of feasible optimizers are feasible for the same combination of their bounds M . This proves concavity. Homogeneity is immediate and continuity follows from the square-root formula.

Apply the result to $M = A \otimes B$. If U is a diagonal unitary, U commutes with A and $(U \otimes U)S = S(U \otimes U)$. Thus simultaneous conjugation shows that the value at UBU^* equals that at B . Averaging diagonal sign unitaries eliminates every off-diagonal entry, and concavity proves the pinching assertion. Pinching may increase rank; it is used below only in an *unrestricted* upper bound. A rank-two optimizer is subsequently constructed, rather than inferred from a rank-preserving pinching. $\square$

10.2. The exact rank-two leaf.

Lemma 10.3 (A centered swap with a rank-two old Gram). *Let $A \succeq 0$ have rank at most two and trace $a > 0$. Let $x, 1 - x$ be the eigenvalues of A/a on a two-dimensional support, padded if necessary, with $1/2 \leq x \leq 1$. For every $\ell \geq 2$,*

$$(10.5) \quad \max_{B \in \mathcal{D}_d^{\leq \ell}} \|(\sqrt{A} \otimes \sqrt{B})(I - dS)(\sqrt{A} \otimes \sqrt{B})\|_1 = a\psi_d(4x(1-x)).$$

The same maximum holds without a rank restriction on B . It has an optimizer commuting with A and supported on its leading two coordinates. For $v = 4x(1-x) > 0$, put

$$(10.6) \quad z = 2x - 1, \quad e = d - 1, \quad T = 2\psi_d(v) - e, \quad w_* = \frac{z(eT - 1)}{c_d(v)}.$$

The two fresh eigenvalues are $(1+w_)/2, (1-w_*)/2$ in that ordered eigenbasis. Here $0 \leq w_* \leq z < 1$. At rank one, an aligned pure fresh Gram is optimal and the value is $a(d-1)$. At $A = 0$ the value is zero.*

Proof. Divide by a and diagonalize A . Lemma 10.2 permits a diagonal B for the unrestricted maximum. If A has rank one, the norm is $(d-1)b_1 + \sum_{j>1} b_j \leq d-1$, attained by $b_1 = 1$. For rank two, write $m = b_1 + b_2$. The part with the fresh coordinate outside the support of A is positive and has trace $1 - m$; the swap term vanishes there. The remaining part has norm m times its normalized two-coordinate value, when $m > 0$. The two-coordinate maximum is at least $d-1$: for example $B = A$ gives

$$(d-1)(x^2 + (1-x)^2) + 2dx(1-x) = d-1 + 2x(1-x).$$

It follows that all fresh mass may be placed in those two coordinates. This also constructs a candidate of rank at most two and justifies applying the unrestricted upper at every $\ell \geq 2$.

Write the two fresh weights as $(1+w)/2, (1-w)/2$, $-1 \leq w \leq 1$. The vectors $|11\rangle, |22\rangle$ give the negative diagonal entries $(1-d)x(1+w)/2$ and $(1-d)(1-x)(1-w)/2$. On $|12\rangle, |21\rangle$ the block is

$$\begin{pmatrix} \frac{x(1-w)/2}{-\frac{d}{2}\sqrt{x(1-x)(1-w^2)}} & -\frac{d}{2}\sqrt{x(1-x)(1-w^2)} \\ -\frac{d}{2}\sqrt{x(1-x)(1-w^2)} & \frac{(1-x)(1+w)/2}{(1-x)(1+w)/2} \end{pmatrix}.$$

Its determinant is nonpositive. Hence the complete trace norm is

$$(10.7) \quad F_z(w) = \frac{1}{2} \left[e(1+zw) + \sqrt{(z-w)^2 + d^2v(1-w^2)} \right], \quad v = 1 - z^2.$$

This formula includes $w = \pm 1$ by continuity.

Here is an algebraic upper certificate, which also evaluates the maximum. Set $C = c_d(v) > 0$ and $D(w) = 1 + (d^2 - 1)v - 2zw + (1 - d^2v)w^2$. Substitution of (10.1)–(10.6) gives

$$(10.8) \quad (T - ezw)^2 - D(w) = C(w - w_*)^2.$$

For clarity, its quadratic coefficient is $e^2z^2 - (1 - d^2v) = C$, its linear coefficient is $-2z(eT - 1)$, and its constant coefficient equals $z^2(eT - 1)^2/C$. The last identity follows by squaring $\sqrt{C/(d^2 - 1)}$ in the expression for T . Since $T \geq e$ and $z < 1$, $T - ezw > 0$ throughout $[-1, 1]$. Moreover

$$eT - 1 = d(d-2) + \frac{e(2d-1)v}{e + d\sqrt{C/(d^2-1)}} \in [0, C],$$

which proves $0 \leq w_* \leq z < 1$. Taking square roots in (10.8) proves $F_z(w) \leq (e + T)/2 = \psi_d(v)$, with equality at w_* . No squared stationary equation with an unchecked sign is used. The rank-one and zero cases were treated before division by C . Finally polar isometries preserve these trace norms for all rectangular coefficient maps having Grams A, B . $\square$

Lemma 10.4 (Concavity of the rank-two spectral objective). *The function ψ_d is continuous, strictly increasing and concave on $[0, 1]$, with*

$$\psi_d(0) = d - 1, \quad \psi_d(1) = d - 1/2.$$

The function $f_d(x, 1 - x) = \psi_d(4x(1 - x))$ is symmetric and strictly concave on the two-coordinate probability simplex.

Proof. In the positive added term of (10.1) write

$$a = d - 1, \quad b = d/\sqrt{d^2 - 1}, \quad c = d(d - 2), \quad s = 2d - 1.$$

It suffices to examine $h(v) = v/(a + b\sqrt{c + sv})$. For $y = \sqrt{c + sv} > 0$ differentiation gives

$$h'(v) = \frac{a + (b/2)(y + c/y)}{(a + by)^2} > 0.$$

The derivative of this last expression with respect to y is

$$-\frac{b}{2(a + by)^3} \left(3a + \frac{ac}{y^2} + by + \frac{3bc}{y} \right) < 0.$$

Since y increases with v , this proves concavity in the interior. When $d = 2$ the formula extends continuously to $v = 0$ with finite right derivative; concavity on the closed interval follows by limits. The endpoint values follow by substitution. Finally $4x(1 - x)$ is strictly concave and symmetric, and composition with an increasing concave function gives the claimed strict concavity of f_d . $\square$

10.3. Full protocol optimization and attainment.

Proof of Theorem 10.1. Write $\alpha = t^2/[2d^2(d + 1)]$. The two equal-prior payoff blocks are those of (5.3): $\alpha(I - dS)$ for equal labels and $-\alpha(I - dS)/(d - 1)$ for unequal labels. Use the dimension-preserving normal form of Lemma 6.2, on the prescribed initial acquisition. For each first label, its Grams satisfy

$$\sum_h A_{yh} = \rho, \quad \text{rank } A_{yh} \leq 2, \quad \text{tr } b_{yh} = 1, \quad \text{rank } b_{yh} \leq \ell.$$

On a nonzero branch put $w_{yh} = \text{tr } A_{yh}$ and $a_{yh} = A_{yh}/w_{yh}$. These are Gram weights, not conditional history probabilities. Optimizing final effects over each second label gives exactly α times the trace norm of the centered filtered swap: the $d - 1$ opposite-sign blocks cancel their denominator, and $\text{tr } X_+ + \text{tr } (-X)_+ = \|X\|_1$. Lemma 10.3 therefore bounds the gain by

$$\alpha \sum_{y,h} w_{yh} f_d(\lambda(a_{yh})).$$

Apply Lemma 8.5 with rank two and the continuous symmetric concave function of Lemma 10.4. For every y the corresponding sum is at most $f_d(q, 1 - q) = \psi_d(v_\rho)$. There are d first labels. This proves the upper in (10.2). Kraus refinements of mixed receiver maps and classical refinements of fresh mixtures do not retain an extra quantum environment. Countable outcomes and null histories use Lemma 2.2; the bound then passes to tester closure.

For attainment choose the at-most- d commuting atoms of Lemma 8.5, $\rho = \sum_h w_h a_h$, all of spectrum $(q, 1 - q, 0, \dots, 0)$. Factor $w_h a_h = C_h^* C_h$ with $C_h : \mathbb{C}^d \rightarrow \mathbb{C}^2$. The maps $V_h = C_h C_0^+$ satisfy $V_h C_0 = C_h$ and $\sum_h V_h^* V_h = I$ on the initial Schmidt support, and extend to a complete recorded instrument on the initial receiver. For each h , choose the commuting maximizing b_h of Lemma 10.3 and a coefficient map $D_h : \mathbb{C}^d \rightarrow \mathbb{C}^2$ with $D_h^* D_h = b_h$. Prepare its normalized vector independently of the old output, conditional on h . For the centered receiver operator X_h , declare C on its positive spectral subspace when labels agree, and on its negative spectral subspace when they differ; assign the kernel arbitrarily. The complementary effects complete a binary POVM

on every block. Every leaf bound and the envelope bound is now attained. The instrument is common to all first labels and is defined without dividing by any physical branch probability.

The endpoint and strictness claims follow from Lemma 10.4. For the rank-one assertion, Theorem 6.5 gives the upper independently of ρ . Decompose this fixed Gram into pure spectral atoms and construct the same support-inverse recorded instrument. Prepare an aligned pure fresh input for each atom. Its centered norm is $d - 1$, so the positive-part decisions attain that upper with both receivers of dimension one. This proves the assertion whenever either of the allowed dimensions is one. The one-call channel identity and independent-redraw conclusion follow from the first moment, exactly as in Theorem 5.2. $\square$

10.4. A complete qubit formula and spectral inference.

Corollary 10.5 (Every pure initial qubit acquisition). *For $d = 2$ and any $\rho \in \mathcal{D}_2$, put $u = \sqrt{\det \rho}$. With both retained registers allowed dimension two, the exact equal-prior reset value is*

$$(10.9) \quad P_{2,2}^{\text{eq}}(\rho) = \frac{1}{2} + \frac{t^2}{12} \left(1 + \frac{6u^2}{1+4u} \right).$$

After identifying the initial Schmidt support with a subspace of $\mathbb{C}^2$, no nontrivial recorded receiver instrument or receiver message is required. In an eigenbasis of ρ , write its eigenvalues as $(1+z)/2, (1-z)/2, 0 \leq z \leq 1$. An optimal fresh Gram has weights

$$(10.10) \quad \frac{1}{2} \left(1 + \frac{z}{1+2\sqrt{1-z^2}} \right), \quad \frac{1}{2} \left(1 - \frac{z}{1+2\sqrt{1-z^2}} \right).$$

The final joint spectral decisions are as in the theorem. All receiver feedback, public randomization and intermediate receiver processing are included in the upper bound.

Proof. Here the initial rank is at most two, so its rank-two majorant is its own spectrum. In the preceding attainment the single atom ρ suffices: keep the first receiver unchanged on that Schmidt support. In (10.1), $v_\rho = 4 \det \rho = 4u^2$ and $\psi_2(v) = 1 + 3v/[2(1+2\sqrt{v})]$. This gives (10.9). For $v > 0$, (10.6) simplifies to $w_* = z/(1+2\sqrt{v})$; its endpoint limit gives the stated pure fresh Gram at $z = 1$. $\square$

Thus a pure initial Gram gives $1/2 + t^2/12$ and the maximally mixed Gram gives $1/2 + t^2/8$. At a strictly intermediate spectrum the maximizing fresh spectrum is generally neither the initial spectrum nor the uniform spectrum. This differs from the biased rank-two leaf, whose optimal fresh weights are equal.

Corollary 10.6 (Exact loss and a calibrated spectral bound). *In Theorem 10.1, the exact loss from the optimized rank-two value is*

$$(10.11) \quad \frac{t^2}{2d(d+1)} \left(d - \frac{1}{2} - \psi_d(v_\rho) \right).$$

Suppose $t > 0$ and an implemented score p differs by at most $\eta \geq 0$ from the score of a nominal strategy with this fixed initial acquisition and the stated reset dimensions. Put

$$R = \frac{2d(d+1)}{t^2} \left(p - \eta - \frac{1}{2} \right).$$

If $d - 1 < R \leq d - 1/2$, let $v_R \in (0, 1]$ be the unique solution $\psi_d(v_R) = R$. Then

$$(10.12) \quad \lambda_{\max}(\rho) \leq \frac{1 + \sqrt{1 - v_R}}{2}.$$

If $R \leq d - 1$ this reasoning gives only the trivial bound one. If $R > d - 1/2$, the nominal model and the supplied error bound cannot both describe that score.

Proof. Subtract (10.2) from (6.8) at rank two. For the second assertion, $p - \eta \leq P_{2,\ell}^{\text{eq}}(\rho)$ gives $R \leq \psi_d(v_\rho)$. Strict monotonicity gives $v_\rho \geq v_R$, and solving $4q(1 - q) \geq v_R$ for $q \geq 1/2$ proves (10.12). The two remaining cases follow from the endpoint range of ψ_d . $\square$

The error hypothesis concerns a specified implementation and nominal experiment, with fixed priors and latent-index law. A uniform unhalved diamond error δ on each complete device channel contributes at most δ to a two-call decision score; a certified hard-path preparation error ε contributes at most $\varepsilon/2$. Their sum may therefore be used for η , as in Corollary 5.3. A fidelity estimate, a sampled matrix inequality or an uncalibrated score is not substituted for this hypothesis. The bound infers a property of the prescribed initial Gram, not its eigenbasis or the dimension of an unrestricted coherent machine.

11. COMPARISON OF OPERATIONAL MODELS

The feasible set, not merely the word “memory”, determines a decision problem. We record four comparisons relevant to the general and spectral results above.

Constrained-separability testers. Ohst, Zhang, Nguyen, Plávala and Quintino [2, Sections 5.3–5.4 and 6.1–6.2] formulate auxiliary memory restrictions by positive separable decompositions with affine conditions on their factors. Their Definition 23 and Theorem 24 characterize complete measure-and-reprepare adaptive testers and their constrained-separability formulation. Our class $\mathfrak{C}$ uses that complete-readout boundary. A reset tester retaining an old receiver instead need not have separable final complete-call effects: the later joint readout is unrestricted, while coherent import into the new source is prohibited. Our fixed-initial objective constrains the common Gram barycenter and evaluates it explicitly for the shared-basis payoff. The general tester framework is an antecedent, not a claim superseded by a first construction of memory hierarchies.

Autonomous coherent memory. Zonnios and Binder [3, Proposition 2, Theorem 1 and Corollary 1] optimize over a repeated stationary instrument carrying a coherent register between steps and retaining the classical record. Their finite-time completeness compiles a general adaptive tester into an autonomous machine with sufficiently large memory and a counter. Here the source cannot receive the old coherent register, public controls may depend on the step, and the two specified receiver cuts are not an all-time workspace dimension. Saturation at rank ρ in our biased experiment concerns the exact value of that family with its first acquisition fixed. It is not their arbitrary-horizon compilation theorem.

Majorization and ensemble transformations. Nielsen [10] characterizes deterministic pure-state transformations by majorization. Jonathan and Plenio [11, Theorem 1] treat ensembles of pure-state outputs through averaged Schmidt constraints. These are direct antecedents for the spectral barycenter step. Our feasible instruments act only on the already prepared receiver, have the common sum $\sum_h A_{yh} = \rho$ after each first device label, and optimize subsequent measurement decisions. Their transformation targets are not the payoff matrices (2.3). The least-majorant argument, finite convexity and support inverse are used here with credit; the specific contribution is their combination with the exact rank-constrained filtered-swap leaf and its message comparator. Assistance and concave roofs. Entanglement assistance [12] and general roof methods [7] optimize averages over decompositions of a fixed state. The present all-density and pure-atom roofs have such a mathematical form, but their atoms, rank cuts and continuation reward come from the reset normal form. The uniform message criterion asks whether these continuation functions are concave; the binary witness changes one receiver-to-source classical permission, not an entanglement-conversion task. A universal majorant is a condition on all admissible atoms and cannot be replaced by finite sampling.

Finally, a measurement channel with classical output is not a Lüders instrument exposing a post-measurement state [4]. The local geometry in the supplement, the general variational

theory, and the closed shared-basis formulas remain separate theorem families. No priority for majorization, concave envelopes, swap tests, or memory-constrained discrimination as general mechanisms is claimed.

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